Cuprins

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Linear Algebra — Solutions & Theory

5 November 2026 · Răzvan

How this document is organized. Each chapter has two parts: *theory* (definitions, theorems with **complete proofs**, a "what to check" recipe for every definition you will have to verify in exercises) and *solutions* of the seminar sheet with the same number. The grey label next to a theorem says where it comes from: **C0x** = course slides, **book** = Crivei, *Basic Linear Algebra* (2022). "Proof: course" means the proof is on the slides; "proof: book" means the slides only state the result and the proof follows the book; "proof: homework" means the slides leave it as homework and it is written out here. All proofs are written step by step, with the reason for each step. Chapter 8 answers Crivei's true/false quizzes and Chapter 9 solves the proof-practice exercises from the workbook. The last part, *Exam proofs — quick sheet*, compresses every proof to its key steps for revision before the partial exam.

Chapter 1 — Operations, monoids, groups: theory

Course 2 (first part) · book §1.4–1.5. Seminar 1 uses only this part.

Operations and stable subsets

Definition (operation). An *operation* (composition law) on a set A is a function $\varphi : A \times A \to A$. We write $x \cdot y$, $x + y$, $x * y$ instead of $\varphi(x, y)$.

What to check. "$*$ is an operation on A" means: for *every* $x, y \in A$, the result $x * y$ is defined and belongs to A. To disprove it, give one pair with $x * y \notin A$ (or undefined).

Definition (stable subset). Let φ be an operation on A and $B \subseteq A$. B is a *stable* (closed) subset of (A, φ) if $\forall x, y \in B$, $\varphi(x, y) \in B$. Then φ induces an operation on B.

Key idea. Laws that use only "$\forall$" (associativity, commutativity, distributivity) automatically pass to a stable subset. Laws that use "$\exists$" (identity element, inverses) do **not** pass automatically — they must be checked again.

Laws of an operation

Definition. Let "$\cdot$" be an operation on A.
(1) associative: $(x \cdot y) \cdot z = x \cdot (y \cdot z)$, $\forall x, y, z \in A$;
(2) commutative: $x \cdot y = y \cdot x$, $\forall x, y \in A$;
(3) identity element: $\exists e \in A$ with $a \cdot e = e \cdot a = a$, $\forall a \in A$;
(4) inverse elements: $\forall a \in A$, $\exists a' \in A$ with $a \cdot a' = a' \cdot a = e$.

Lemma (uniqueness of identity and inverse). C02 · proof: book Lemma 1.4.4
(i) If an operation has an identity element, it is unique.
(ii) If "$\cdot$" is associative with identity e and a has an inverse, then this inverse is unique.

Proof. (i) Let e_1, e_2 be identity elements. Compute $e_1 \cdot e_2$ in two ways:

$$e_1 \cdot e_2 = e_1 \ \text{(because } e_2 \text{ is an identity)}, \qquad e_1 \cdot e_2 = e_2 \ \text{(because } e_1 \text{ is an identity)}.$$

Hence $e_1 = e_2$.

(ii) Let a_1, a_2 be inverses of a. Using associativity, compute $a_1 \cdot a \cdot a_2$ in two ways:

$$a_1 \cdot (a \cdot a_2) = a_1 \cdot e = a_1, \qquad (a_1 \cdot a) \cdot a_2 = e \cdot a_2 = a_2.$$

Hence $a_1 = a_2$. ∎

Key idea. Compute one product in two ways. (ii) needs associativity — this is the only place where it is used.

Semigroups, monoids, groups

Definition. $(A, \cdot)$ is a *semigroup* if "$\cdot$" is associative; a *monoid* if it is a semigroup with identity element; a *group* if it is a monoid in which every element has an inverse. If moreover "$\cdot$" is commutative, the structure is *commutative*; a commutative group is called *abelian*. Notation: in a group $(G, \cdot)$ the identity is 1 and the inverse of x is x^{-1}; in $(G, +)$ they are 0 and $-x$ (the symmetric of x).

What to check — "$(A, *)$ is a group".

1. $*$ is an operation on A (closure);
2. associativity — compute both sides for arbitrary x, y, z;
3. identity: *find* a candidate e (solve $x * e = x$), then verify $x * e = e * x = x$ for all x;
4. inverses: for an arbitrary x, *solve* $x * x' = e$, check $x' \in A$ and $x' * x = e$.

For a monoid stop after step 3; "commutative" adds $x * y = y * x$.

Definition (powers). In a semigroup, $x^n = x \cdot x \cdots x$ (n times, $n \in \mathbb{N}^*$); in a monoid also $x^0 = 1$; in a group also $x^{-n} = (x^{-1})^n$. In additive notation x^n becomes nx.

Lemma (power rules). C02 · proof: book Lemma 1.5.5 In a group $(G, \cdot)$: $x^m x^n = x^{m+n}$ and $(x^m)^n = x^{mn}$, $\forall x \in G$, $\forall m, n \in \mathbb{Z}$.

Proof. For $m, n \in \mathbb{N}$, $x^m x^n$ is a product of $m + n$ factors x (associativity lets us drop the brackets), so it equals x^{m+n}; formally, induction on n: $n = 0$ is $x^m \cdot 1 = x^m$, and $x^m x^{n+1} = (x^m x^n)x = x^{m+n}x = x^{m+n+1}$. Similarly $(x^m)^{n+1} = (x^m)^n x^m = x^{mn} x^m = x^{mn+m}$. For negative exponents write $x^{-k} = (x^{-1})^k$ and use $x \cdot x^{-1} = 1$ to cancel factors pairwise; e.g. for $m \geq k \geq 0$, $x^m x^{-k} = x^{m-k}(x^k x^{-k}) = x^{m-k}$, since $x^k (x^{-1})^k = 1$ by repeatedly cancelling the middle pair $x\, x^{-1}$. ∎

Lemma (computation in groups). C02 · proof: book Lemma 1.5.6 (written out) Let $(G, \cdot)$ be a group and $a, x, y \in G$. Then:
(i) $ax = ay \Rightarrow x = y$ and $xa = ya \Rightarrow x = y$ (cancellation laws);
(ii) $(x^{-1})^{-1} = x$; (iii) $(xy)^{-1} = y^{-1}x^{-1}$.

Proof. (i) Multiply $ax = ay$ on the left by a^{-1}: $a^{-1}(ax) = a^{-1}(ay) \Rightarrow (a^{-1}a)x = (a^{-1}a)y \Rightarrow 1 \cdot x = 1 \cdot y \Rightarrow x = y$. The right law: multiply on the right by a^{-1}.

(ii) $x \cdot x^{-1} = x^{-1} \cdot x = 1$ says exactly that x is an inverse of x^{-1}. By uniqueness of the inverse, $(x^{-1})^{-1} = x$.

(iii) $(xy)(y^{-1}x^{-1}) = x(yy^{-1})x^{-1} = x\,1\,x^{-1} = 1$ and $(y^{-1}x^{-1})(xy) = y^{-1}(x^{-1}x)y = y^{-1}y = 1$. So $y^{-1}x^{-1}$ is an inverse of xy; by uniqueness it is $(xy)^{-1}$. ∎

Key idea. To prove "b is the inverse of a", show $ab = ba = 1$ and invoke uniqueness. Cancellation ⇒ in the operation table of a finite group every element appears exactly once in each row and column.

Standard examples

Structure	Type	Why
$(\mathbb{N}^*, +)$	semigroup, not monoid	$0 \notin \mathbb{N}^*$
$(\mathbb{N}, +)$; $(\mathbb{N}, \cdot)$, $(\mathbb{Z}, \cdot)$, $(\mathbb{Q}, \cdot)$, $(\mathbb{R}, \cdot)$, $(\mathbb{C}, \cdot)$	monoids, not groups	1 has no symmetric in $\mathbb{N}$; 0 has no inverse
$(\mathbb{Z}, +)$, $(\mathbb{Q}, +)$, $(\mathbb{R}, +)$, $(\mathbb{C}, +)$, $(\mathbb{Q}^*, \cdot)$, $(\mathbb{R}^*, \cdot)$, $(\mathbb{C}^*, \cdot)$	abelian groups	
$(\mathbb{Z}_n, +)$, $\widehat{x} + \widehat{y} = \widehat{x + y}$	abelian group	identity $\widehat{0}$, $-\widehat{x} = \widehat{n - x}$
$(S_M, \circ)$, $(\mathrm{GL}_n(\mathbb{R}), \cdot)$	groups, not abelian (for $\lvert M\rvert \geq 3$, $n \geq 2$)	Seminar 3
Klein's group $K = \{e, a, b, c\}$	abelian group, $x^2 = e$ for all x	symmetries of a rectangle
$(X^*, \cdot)$ words with concatenation	monoid (free monoid), identity = void word	

Chapter 1 — Solutions (Seminar 1)

Ex. 1

An operation must give a result *in the same set* for *every* pair.

	$\mathbb{N}$	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{R}$	$\mathbb{C}$
$+$	yes	yes	yes	yes	yes
$-$	no: $1-2=-1\notin\mathbb{N}$	yes	yes	yes	yes
$\cdot$	yes	yes	yes	yes	yes
$:$	no	no	no	no	no

Division is never an operation on these sets: $x:0$ is not defined, so it is not defined on all of $A\times A$ (also $1:2\notin\mathbb{N},\mathbb{Z}$). Division *is* an operation on $\mathbb{Q}^*$, $\mathbb{R}^*$, $\mathbb{C}^*$.

Ex. 2

An operation on $A=\{a_1,a_2,a_3\}$ is the same as its operation table: $3\times 3=9$ cells, each filled with one of the 3 elements.

(i) Each cell independently: $3^9=19683$ operations. (Equivalently, functions $A\times A\to A$: $|A|^{|A\times A|}=3^9$.)

(ii) Commutative $\iff$ the table is symmetric with respect to the main diagonal. Free cells: the 3 diagonal cells and the 3 cells above the diagonal; the others are copies. $\implies 3^6=729$.

(iii) Choose the identity e: 3 ways. Its row and column are forced ($e\cdot x=x\cdot e=x$): $2\cdot 3-1=5$ cells. The remaining $(3-1)^2=4$ cells are free: 3^4 ways. An identity is unique (Lemma), so no table is counted twice $\implies 3\cdot 3^4=243$.

General $|A|=n$:

$$\#\text{operations}=n^{n^2},\qquad \#\text{commutative}=n^{\frac{n(n+1)}{2}},\qquad \#\text{with identity}=n\cdot n^{(n-1)^2}=n^{n^2-2n+2}.$$

Check with $n=3$: 3^9, 3^6, $3^5=243$. ✓

Ex. 3

	with $+$	**with** $\cdot$
$\mathbb{N}$	monoid, not group: 1 has no symmetric ($1+x=0$ has no solution in $\mathbb{N}$)	monoid, not group: 0 (and 2) have no inverse
$\mathbb{Z}$	group (abelian)	monoid, not group: $2x=1$ has no solution in $\mathbb{Z}$
$\mathbb{Q},\mathbb{R},\mathbb{C}$	groups (abelian)	monoids, not groups: 0 has no inverse

Removing 0: $(\mathbb{Q}^*,\cdot)$, $(\mathbb{R}^*,\cdot)$, $(\mathbb{C}^*,\cdot)$ are abelian groups; $(\mathbb{Z}^*,\cdot)$ is still not a group.

Ex. 4

Key observation: $x*y=x+y+xy=(1+x)(1+y)-1$.

(i) *Operation:* $x+y+xy\in\mathbb{R}$ for all $x,y\in\mathbb{R}$.

Commutative: $x*y=(1+x)(1+y)-1=(1+y)(1+x)-1=y*x$.

Associative: $1 + (x * y) = (1 + x)(1 + y)$, so

$$(x * y) * z = \big(1 + x * y\big)(1 + z) - 1 = (1 + x)(1 + y)(1 + z) - 1,$$
$$x * (y * z) = (1 + x)\big(1 + y * z\big) - 1 = (1 + x)(1 + y)(1 + z) - 1.$$

The two are equal.

Identity: we look for e with $x * e = x$ for all x: $x + e + xe = x \iff e(1 + x) = 0$ for all $x \implies e = 0$. Check: $x * 0 = x + 0 + 0 = x$ and $0 * x = x$ (commutativity).

$\implies (\mathbb{R}, *)$ **is a commutative monoid with identity element** 0**.**

(ii) Let $x, y \in [-1, \infty)$. Then $1 + x \geq 0$ and $1 + y \geq 0 \implies (1 + x)(1 + y) \geq 0 \implies x * y = (1 + x)(1 + y) - 1 \geq -1$.

$\implies x * y \in [-1, \infty)$**, so** $[-1, \infty)$ **is a stable subset of** $(\mathbb{R}, *)$**.**

Remark: $(\mathbb{R}, *)$ is not a group: $-1 * y = -1$ for every y, so -1 has no inverse. In fact $(\mathbb{R} \setminus \{-1\}, *)$ is a group: the inverse of x is $-\frac{x}{1+x}$.

Ex. 5

Here $\gcd(x, y)$ is the greatest common divisor with respect to divisibility, with $\gcd(0, 0) = 0$; recall that every x divides 0.

(i) *Operation:* $\gcd(x, y) \in \mathbb{N}$. *Commutative:* the common divisors of x, y are those of y, x.

Associative: for every $d \in \mathbb{N}$,

$$d \mid \gcd(\gcd(x, y), z) \iff d \mid \gcd(x, y) \text{ and } d \mid z \iff d \mid x,\ d \mid y,\ d \mid z,$$

and in the same way $d \mid \gcd(x, \gcd(y, z)) \iff d \mid x,\ d \mid y,\ d \mid z$. Two natural numbers with exactly the same divisors are equal (each divides the other) $\implies (x * y) * z = x * (y * z)$.

Identity: $x * 0 = \gcd(x, 0) = x$, because $x \mid x$ and $x \mid 0$ and any common divisor divides x. So $e = 0$.

$\implies (\mathbb{N}, *)$ **is a commutative monoid with identity** 0**.**

(ii) Let $x, y \in D_n$. Then $\gcd(x, y) \mid x$ and $x \mid n \implies \gcd(x, y) \mid n \implies x * y \in D_n$: D_n is stable. Associativity and commutativity pass to the stable subset. Identity: $0 \notin D_n$ (as $n \neq 0$), but $n \in D_n$ and $x * n = \gcd(x, n) = x$ for every $x \mid n$.

$\implies (D_n, *)$ **is a commutative monoid with identity** n**.**

(iii) $D_6 = \{1, 2, 3, 6\}$:

$*$	**1**	**2**	**3**	**6**
1	1	1	1	1
2	1	2	1	2
3	1	1	3	3
6	1	2	3	6

The row of 6 copies the header: 6 is the identity. The table is symmetric: the operation is commutative.

Ex. 6

Let S be a finite stable subset of $(\mathbb{Z}, \cdot)$ and $x \in S$. By stability, $x^2, x^3, \cdots \in S$.

If $|x| \geq 2$, then $|x| < |x|^2 < |x|^3 < \cdots$, so the powers x^k are pairwise distinct $\Longrightarrow$ S infinite, contradiction. Hence $S \subseteq \{-1, 0, 1\}$.

Checking the subsets of $\{-1, 0, 1\}$ (only products of elements of S matter):

- stable: $\emptyset$, $\{0\}$, $\{1\}$, $\{0, 1\}$, $\{-1, 1\}$, $\{-1, 0, 1\}$;
- not stable: $\{-1\}$ and $\{-1, 0\}$, because $(-1)(-1) = 1 \notin S$.

$\Longrightarrow$ **The finite stable subsets are** $\emptyset, \{0\}, \{1\}, \{0, 1\}, \{-1, 1\}, \{-1, 0, 1\}$**.**

Ex. 7

(i) "$\Longrightarrow$" If G is abelian, then $(xy)^2 = xyxy = x(yx)y = x(xy)y = x^2y^2$ (associativity, then $yx = xy$).

"$\Longleftarrow$" Assume $(xy)^2 = x^2y^2$, i.e. $x\,y\,x\,y = x\,x\,y\,y$. Cancel x on the left and y on the right (cancellation laws) $\Longrightarrow$ $yx = xy$. As x, y were arbitrary, G is abelian.

(ii) $x^2 = 1$ means $x \cdot x = 1$, i.e. $x^{-1} = x$ for every $x \in G$. Then, for all $x, y \in G$:

$$xy = (xy)^{-1} = y^{-1}x^{-1} = yx.$$

$\Longrightarrow$ G **is abelian. (Example: Klein's group.)**

Alternative for (ii) using (i): $(xy)^2 = 1 = 1 \cdot 1 = x^2y^2$.

Ex. 8

(i) *Operation:* $X * Y \subseteq A$, so $X * Y \in \mathcal{P}(A)$.

Associative: using the associativity of "$\cdot$",

$$(X * Y) * Z = \{(xy)z \mid x \in X, y \in Y, z \in Z\} = \{x(yz) \mid x \in X, y \in Y, z \in Z\} = X * (Y * Z).$$

Identity: let e be the identity of $(A, \cdot)$. Then $X * \{e\} = \{xe \mid x \in X\} = X$ and $\{e\} * X = X$.

$\Longrightarrow$ $(\mathcal{P}(A), *)$ **is a monoid with identity** $\{e\}$**.**

(ii) Take $X = \emptyset$. For every Y, $\emptyset * Y = \emptyset \neq \{e\}$ (there is no $x \in \emptyset$). So $\emptyset$ has no inverse.

$\Longrightarrow$ $(\mathcal{P}(A), *)$ **is not a group (for any group** A**).**

Another obstruction, if $|A| \geq 2$: for $Y \neq \emptyset$, $A * Y = A$ (any $a = (ay^{-1})y$), so $A * Y \neq \{e\}$ and A has no inverse either.

Results at a glance — Seminar 1

Ex.	Answer
1	$+, \cdot$: operations on all five sets; $-$: not on $\mathbb{N}$; division: on none
2	$3^9 = 19683$; $3^6 = 729$; 243; in general n^{n^2}, $n^{n(n+1)/2}$, n^{n^2-2n+2}
3	groups: $(\mathbb{Z}, +), (\mathbb{Q}, +), (\mathbb{R}, +), (\mathbb{C}, +)$; all others are monoids only
4	identity 0; key: $x * y = (1 + x)(1 + y) - 1$

5	identity of $(\mathbb{N}, \gcd)$ is 0; of $(D_n, \gcd)$ is n
6	$\emptyset, \{0\}, \{1\}, \{0,1\}, \{-1,1\}, \{-1,0,1\}$
7	cancellation; $x^{-1} = x$
8	identity $\{e\}$; $\emptyset$ has no inverse

Chapter 2 — Relations, functions, equivalences: theory

Course 1 · book §1.1–1.3.

Relations and functions

Definition (relation). A (binary) relation is a triple $r = (A, B, R)$, where A, B are sets and $R \subseteq A \times B$. A = domain, B = codomain, R = graph. We write $a\,r\,b$ for $(a,b) \in R$. If $A = B$, r is *homogeneous*. For $X \subseteq A$, the *relation class* of X is $r(X) = \{b \in B \mid \exists x \in X : x\,r\,b\}$, and $r\langle x\rangle = r(\{x\})$. Note $r(X) = \bigcup_{x\in X} r\langle x\rangle$.

Definition (function). A relation $r = (A, B, R)$ is a *function* if $|r\langle a\rangle| = 1$ for every $a \in A$: every element of the domain is related to **exactly one** element of the codomain, denoted $f(a)$. The image of $X \subseteq A$ is $f(X) = \{f(x) \mid x \in X\}$, and $\operatorname{Im} f = f(A)$.

What to check — "h is a function". For every x in the domain: (1) *existence*: some y with $x\,h\,y$; (2) *uniqueness*: if $x\,h\,y_1$ and $x\,h\,y_2$ then $y_1 = y_2$.

Definition. $f : A \to B$ is *injective* if $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$; *surjective* if $\forall y \in B\ \exists x \in A : f(x) = y$ (i.e. $f(A) = B$); *bijective* if both. $g : B \to A$ is an *inverse* of f if $g \circ f = 1_A$ and $f \circ g = 1_B$.

Lemma (inverse function). book Lemma 1.2.10 · proof written out $f : A \to B$ has an inverse $\iff$ f is bijective. The inverse is then unique, denoted f^{-1}.

Proof. "$\Longrightarrow$" Let g be an inverse. *Injective:* $f(x_1) = f(x_2) \Rightarrow g(f(x_1)) = g(f(x_2)) \Rightarrow x_1 = x_2$, since $g \circ f = 1_A$. *Surjective:* for $y \in B$, $y = 1_B(y) = f(g(y))$, so $y = f(x)$ with $x = g(y)$.

"$\Longleftarrow$" For $y \in B$ there is an x with $f(x) = y$ (surjectivity) and only one (injectivity); define $g(y) = x$. Then $f(g(y)) = y$ and $g(f(x)) = x$, i.e. $f \circ g = 1_B$, $g \circ f = 1_A$.

Uniqueness: if g_1, g_2 are inverses, then $g_1 = g_1 \circ 1_B = g_1 \circ (f \circ g_2) = (g_1 \circ f) \circ g_2 = 1_A \circ g_2 = g_2$. ∎

Theorem (number of functions). book Thm 1.2.4 · proof written out If $|A| = n$, $|B| = m$ ($m, n \in \mathbb{N}^*$), then the set B^A of all functions $A \to B$ has $|B^A| = m^n$.

Proof. Induction on n. For $n = 1$, $A = \{a\}$, and a function is determined by $f(a) \in B$: m functions. Step $n \to n+1$: write $A = A' \cup \{a\}$ with $|A'| = n$, $a \notin A'$. A function $f : A \to B$ is the

same as a pair (restriction $f|_{A'}$, value $f(a)$): m^n choices for the first (induction hypothesis) and m for the second. Total $m^n \cdot m = m^{n+1}$. ∎

Counting recipes. relations $A \to B$ = subsets of $A \times B$: $2^{|A||B|}$; operations on A = functions $A \times A \to A$: $|A|^{|A|^2}$.

Equivalence relations

Definition. A homogeneous relation r on A is *reflexive* (r) if $x\,r\,x$ for all $x \in A$; *transitive* (t) if $x\,r\,y$ and $y\,r\,z \Rightarrow x\,r\,z$; *symmetric* (s) if $x\,r\,y \Rightarrow y\,r\,x$. r is an *equivalence relation* if it has (r), (t) and (s).

What to check — "r is an equivalence". Prove (r), (s), (t) for *arbitrary* elements. Typical trick: if $x\,r\,y$ is defined as "$\varphi(x) = \varphi(y)$" for some function φ (e.g. $|z_1| = |z_2|$), all three properties follow from the properties of equality. To show "not an equivalence", give **one** concrete counterexample to one property.

Definition (partition). A partition of $A \neq \emptyset$ is a set π of **non-empty** subsets of A (blocks) such that the blocks cover A ($\bigcup_{B\in\pi} B = A$) and distinct blocks are disjoint. Equivalently: every element of A lies in exactly one block.

Definition. For an equivalence r on A: $r\langle x\rangle$ is the *equivalence class* of x (x is a representative); $A/r = \{r\langle x\rangle \mid x \in A\}$ is the *quotient set*. For a partition π, the *associated relation* is $x\,r_\pi\,y \iff \exists B \in \pi : x, y \in B$. Notation: $E(A)$ = equivalences on A, $P(A)$ = partitions of A.

Theorem (equivalences ⟷ partitions). C01 · proof: book Thm 1.3.7
(i) If $r \in E(A)$, then $A/r \in P(A)$.
(ii) If $\pi \in P(A)$, then $r_\pi \in E(A)$.
(iii) $F : E(A) \to P(A)$, $F(r) = A/r$ is a bijection with inverse $G : P(A) \to E(A)$, $G(\pi) = r_\pi$.

Proof. (i) *Non-empty blocks and covering:* by (r), $x \in r\langle x\rangle$; so every class is non-empty and $A \subseteq \bigcup_x r\langle x\rangle \subseteq A$.

Disjointness: we show that two classes are either equal or disjoint. Let $a \in r\langle x\rangle \cap r\langle y\rangle$, i.e. $x\,r\,a$ and $y\,r\,a$. By (s), $a\,r\,y$; by (t), $x\,r\,y$. Now if $b \in r\langle y\rangle$ ($y\,r\,b$), then $x\,r\,y$, $y\,r\,b$ give $x\,r\,b$ (t), so $r\langle y\rangle \subseteq r\langle x\rangle$. Symmetrically (using $y\,r\,x$ from (s)) $r\langle x\rangle \subseteq r\langle y\rangle$. Hence intersecting classes are equal, and A/r is a partition.

(ii) (r): x lies in some block (covering), so $x\,r_\pi\,x$. (s): "$x, y \in B$" is symmetric in x, y. (t): let $x, y \in B_1$ and $y, z \in B_2$. Then $y \in B_1 \cap B_2$, so $B_1 = B_2$ (distinct blocks are disjoint) and $x, z \in B_1$, i.e. $x\,r_\pi\,z$.

(iii) We show $G \circ F = 1_{E(A)}$ and $F \circ G = 1_{P(A)}$; then F is bijective with $F^{-1} = G$ (Lemma above).

$G(F(r)) = r$: $x\,G(A/r)\,y \iff \exists a : x, y \in r\langle a\rangle \iff x\,r\,y$. ($\Longleftarrow$: take $a = x$, since $x \in r\langle x\rangle$, $y \in r\langle x\rangle$. $\Longrightarrow$: $a\,r\,x$ and $a\,r\,y$ give $x\,r\,a$ by (s), then $x\,r\,y$ by (t).)

$F(G(\pi)) = \pi$: if $x \in B \in \pi$, then $r_\pi\langle x\rangle = \{y \mid x, y \text{ in the same block}\} = B$. So the classes of r_π are exactly the blocks of π. ∎

Key idea. Classes are either equal or disjoint — this uses (s)+(t). In exercises: "quotient set" = list of blocks; "corresponding equivalence" = all pairs inside the same block, i.e. $R_\pi = \bigcup_{B\in\pi} B \times B$.

Example (congruence modulo n). $x \,\rho_n\, y \iff n \mid x - y$. For $n \geq 2$, $\mathbb{Z}/\rho_n = \mathbb{Z}_n = \{\widehat{0}, \widehat{1}, \ldots, \widehat{n-1}\}$ with $\widehat{x} = x + n\mathbb{Z}$; ρ_0 = equality ($\mathbb{Z}/\rho_0 = \{\{x\} \mid x \in \mathbb{Z}\}$), ρ_1 = universal relation ($\mathbb{Z}/\rho_1 = \{\mathbb{Z}\}$).

Chapter 2 — Solutions (Seminar 2)

Ex. 1

$M = \{2, 3, 4, 5, 6\}$; list all pairs $(x, y) \in M \times M$ with the property.

$R = \{(2,3), (2,4), (2,5), (2,6), (3,4), (3,5), (3,6), (4,5), (4,6), (5,6)\}$ (10 pairs).

$S = \{(2,2), (2,4), (2,6), (3,3), (3,6), (4,4), (5,5), (6,6)\}$ (8 pairs).

$T = \{(2,3), (2,5), (3,2), (3,4), (3,5), (4,3), (4,5), (5,2), (5,3), (5,4), (5,6), (6,5)\}$ (12 pairs; no (x,x), since $\gcd(x,x) = x \geq 2$).

V: the remainders mod 3 are $2 \mapsto 2$, $3 \mapsto 0$, $4 \mapsto 1$, $5 \mapsto 2$, $6 \mapsto 0$, so the classes are $\{2,5\}, \{3,6\}, \{4\}$ and

$V = \{(2,2), (2,5), (5,2), (5,5), (3,3), (3,6), (6,3), (6,6), (4,4)\}$ (9 pairs).

Ex. 2

(i) A relation with domain A and codomain B is determined by its graph $R \subseteq A \times B$, and $|A \times B| = nm$. A set with nm elements has 2^{nm} subsets $\Longrightarrow$ 2^{nm} relations.

(ii) Take $B = A$: 2^{n^2} homogeneous relations.

Ex. 3

All examples on small sets; the check is done pair by pair.

- **only (r):** on $A = \{1,2,3\}$, $R = \Delta_A \cup \{(1,2), (2,3)\}$. Reflexive. Not symmetric: $(1,2) \in R$, $(2,1) \notin R$. Not transitive: $(1,2), (2,3) \in R$, $(1,3) \notin R$.
- **only (t):** the strict inequality "$<$" on $\mathbb{R}$. Transitive. Not reflexive: $1 \not< 1$. Not symmetric: $1 < 2$, $2 \not< 1$.
- **only (s):** on $A = \{1,2\}$, $R = \{(1,2), (2,1)\}$. Symmetric. Not reflexive: $(1,1) \notin R$. Not transitive: $(1,2), (2,1) \in R$ but $(1,1) \notin R$.

Ex. 4

Relation	(r)	(s)	(t)	Counterexamples
$<$ on $\mathbb{R}$	no	no	yes	$1 \not< 1$; $1 < 2$, $2 \not< 1$
divisibility on $\mathbb{N}$	yes	no	yes	$1 \mid 2$, $2 \nmid 1$
divisibility on $\mathbb{Z}$	yes	no	yes	$1 \mid 2$, $2 \nmid 1$ (although $2 \mid -2$ and $-2 \mid 2$)
$\perp$ of lines in space	no	yes	no	$a \not\perp a$; $a \perp b$, $b \perp a$, but $a \not\perp a$
$\parallel$ of lines in space	yes*	yes	yes*	*with the convention "every line is parallel to itself"
congruence of triangles	yes	yes	yes	equivalence
similarity of triangles	yes	yes	yes	equivalence

Proofs of the "yes" entries: for divisibility, $x = 1 \cdot x$; $x \mid y$, $y \mid z \implies y = ax$, $z = by \implies z = (ba)x$. For $\parallel$: if "parallel" means "distinct and parallel", then it is symmetric but neither reflexive nor transitive ($a \parallel b$, $b \parallel a$, but not $a \parallel a$); with the convention $a \parallel a$ it is an equivalence whose classes are the *directions*. Congruence and similarity: every triangle is congruent (similar) to itself, the relations are symmetric, and composing two isometries (similarities) gives an isometry (similarity).

Ex. 5

(i) $R_1 = \Delta_M \cup \{(1,2),(2,1),(1,3),(3,1),(2,3),(3,2)\} = \{1,2,3\} \times \{1,2,3\} \cup \{(4,4)\}$.

So $R_1 = \bigcup_{B \in \pi} B \times B$ for $\pi = \{\{1,2,3\},\{4\}\}$; by the theorem r_1 is an equivalence (directly: (r) $\Delta_M \subseteq R_1$; (s) the added pairs come in pairs $(a,b),(b,a)$; (t) two related elements always lie in the same block).

$\implies$ r_1 **is an equivalence,** $M/r_1 = \{\{1,2,3\},\{4\}\}$**.**

R_2: $(1,2) \in R_2$ but $(2,1) \notin R_2 \implies$ not symmetric.

$\implies$ r_2 **is not an equivalence.**

(ii) π_1: the blocks $\{1\},\{2\},\{3,4\}$ are non-empty, pairwise disjoint and their union is M.

$\implies$ π_1 **is a partition;** $R_{\pi_1} = \Delta_M \cup \{(3,4),(4,3)\}$**.**

π_2: $\{1\} \cap \{1,2\} = \{1\} \neq \emptyset$ (1 lies in two blocks).

$\implies$ π_2 **is not a partition.**

Ex. 6

Relation r**.** $z_1 \, r \, z_2 \iff |z_1| = |z_2|$, an equality of images of the function $z \mapsto |z|$:

- (r) $|z| = |z|$; (s) $|z_1| = |z_2| \Rightarrow |z_2| = |z_1|$; (t) $|z_1| = |z_2|$, $|z_2| = |z_3| \Rightarrow |z_1| = |z_3|$.

Class of z: $r\langle z \rangle = \{w \in \mathbb{C} \mid |w| = |z|\}$ = the circle centered at the origin of radius $|z|$ (for $z = 0$, the single point $\{0\}$).

$\implies \mathbb{C}/r = \{\{0\}\} \cup \{\, \mathcal{C}(0,\rho) \mid \rho > 0 \,\}$ **— all concentric circles around the origin, plus the origin.**

Relation s. $\arg z$ is defined only for $z \neq 0$.

- (r) for $z \neq 0$, $\arg z = \arg z$; for $z = 0$, the second condition holds.
- (s) both conditions are symmetric in z_1, z_2.
- (t) let $z_1 \, s \, z_2$ and $z_2 \, s \, z_3$. If $z_2 = 0$, then $z_1 \, s \, 0$ forces $z_1 = 0$ (the condition with arg cannot hold, since $\arg 0$ is not defined), and similarly $z_3 = 0$, so $z_1 \, s \, z_3$. If $z_2 \neq 0$, then $z_1, z_3 \neq 0$ and $\arg z_1 = \arg z_2 = \arg z_3$.

Class of $z \neq 0$ with $\arg z = \theta$: $\{t(\cos\theta + i \sin\theta) \mid t > 0\}$, the open half-line (ray) from the origin through z; the class of 0 is $\{0\}$.

$\Longrightarrow \mathbb{C}/s = \{\{0\}\} \cup \{\text{open rays from } 0\}$**, one ray for each** $\theta \in [0, 2\pi)$**.**

Ex. 7

(r) $n \mid 0 = x - x$. (s) $n \mid x - y \Rightarrow n \mid -(x - y) = y - x$. (t) $n \mid x - y$ and $n \mid y - z \Longrightarrow n \mid (x - y) + (y - z) = x - z$.

$\Longrightarrow \rho_n$ **is an equivalence relation.**

Class of x: $\rho_n\langle x \rangle = \{y \mid n \mid y - x\} = \{x + nk \mid k \in \mathbb{Z}\} = x + n\mathbb{Z} = \widehat{x}$.

$n \geq 2$**:** by division with remainder, $x = nq + r$ with a unique $r \in \{0, \ldots, n-1\}$, so $\widehat{x} = \widehat{r}$. The classes $\widehat{0}, \ldots, \widehat{n-1}$ are distinct: for $0 \leq r < r' \leq n - 1$, $0 < r' - r < n$, so $n \nmid r' - r$.

$\Longrightarrow \mathbb{Z}/\rho_n = \{\widehat{0}, \widehat{1}, \ldots, \widehat{n-1}\} = \mathbb{Z}_n$ **(n classes).**

$n = 0$**:** $0 \mid x - y \iff x - y = 0$, so ρ_0 is the equality relation, $\mathbb{Z}/\rho_0 = \{\{x\} \mid x \in \mathbb{Z}\}$ (all singletons).

$n = 1$**:** 1 divides everything, $\rho_1 = \mathbb{Z} \times \mathbb{Z}$ is the universal relation, $\mathbb{Z}/\rho_1 = \{\mathbb{Z}\}$ (one block).

Ex. 8

By the theorem, it is enough to list the partitions of $M = \{1, 2, 3\}$; each gives exactly one equivalence ($R_\pi = \bigcup B \times B$).

Partition	Graph of the equivalence relation
$\{\{1\}, \{2\}, \{3\}\}$	Δ_M
$\{\{1, 2\}, \{3\}\}$	$\Delta_M \cup \{(1, 2), (2, 1)\}$
$\{\{1, 3\}, \{2\}\}$	$\Delta_M \cup \{(1, 3), (3, 1)\}$
$\{\{2, 3\}, \{1\}\}$	$\Delta_M \cup \{(2, 3), (3, 2)\}$
$\{\{1, 2, 3\}\}$	$M \times M$

$\Longrightarrow$ **5 partitions and 5 equivalence relations.**

Ex. 9

We need: for every $x \in \mathbb{Z}$, exactly one $y \in M$ with $(x, y) \in H$.

Existence: division with remainder by 4: $x = 4q + r$ with $r \in \{0, 1, 2, 3\} = M$; take $z = q$, $y = r$.

Uniqueness: if $x = 4z_1 + y_1 = 4z_2 + y_2$ with $y_1, y_2 \in M$, then $4 \mid y_1 - y_2$ and $|y_1 - y_2| \leq 3$, hence $y_1 = y_2$.

$\Longrightarrow$ h is a function: $h(x)$ = the remainder of x when divided by 4.

Ex. 10

Relation r: (r) $m = 2^0 m$. (t) $m = 2^a n$, $n = 2^b p \Longrightarrow m = 2^{a+b} p$. (s) fails: $2\, r\, 1$ ($2 = 2^1 \cdot 1$), but $1 = 2^a \cdot 2$ has no solution $a \in \mathbb{N}$.

$\Longrightarrow$ r is not an equivalence relation (it is reflexive and transitive only).

Relation s: (r) $m = m$. (s) the three conditions are symmetric in m, n (the last two swap). (t) fails: $2\, s\, 4$ ($4 = 2^2$) and $4\, s\, 16$ ($16 = 4^2$), but $2 \neq 16$, $2 \neq 16^2$, $16 \neq 2^2$, so 2 is not in relation s with 16 .

$\Longrightarrow$ s is not an equivalence relation.

Results at a glance — Seminar 2

Ex.	Answer
1	$\lvert R\rvert = 10$, $\lvert S\rvert = 8$, $\lvert T\rvert = 12$, $\lvert V\rvert = 9$; classes of v: $\{2,5\}, \{3,6\}, \{4\}$
2	2^{nm}; 2^{n^2}
5	r_1 yes, $\{\{1,2,3\},\{4\}\}$; r_2 no; π_1 yes; π_2 no
6	$\mathbb{C}/r$: circles centered at 0 and $\{0\}$; $\mathbb{C}/s$: open rays from 0 and $\{0\}$
7	$\mathbb{Z}_n$ ($n \geq 2$); singletons ($n = 0$); $\{\mathbb{Z}\}$ ($n = 1$)
8	5 equivalences $\leftrightarrow$ 5 partitions
9	yes, $h(x) = x \bmod 4$
10	r: not symmetric; s: not transitive — neither is an equivalence

Chapter 3 — Groups, rings, subgroups, homomorphisms: theory

Course 2 · book §1.5–1.8.

Rings and fields

> **Definition.** $(R, +, \cdot)$ is a *ring* if $(R, +)$ is an abelian group, $(R, \cdot)$ is a semigroup and the distributive laws hold: $x(y + z) = xy + xz$, $(y + z)x = yx + zx$. It is *unitary* if "$\cdot$" has an identity 1; a *division ring* if $(R, +)$ is an abelian group, $(R^*, \cdot)$ is a group ($R^* = R \setminus \{0\}$) and the distributive laws hold; a *field* if it is a commutative division ring.

> **What to check — "$(R, +, \cdot)$ is a field".** (1) $(R, +)$ abelian group; (2) "$\cdot$" associative and commutative, with identity $1 \neq 0$; (3) every $x \neq 0$ has an inverse; (4) distributivity.

Shortcut: if R is a subset of a known ring closed under the operations, (1), (4) and associativity are inherited (Ex. 10).

Lemma (rules in a ring). C02 · proof: book Lemma 1.5.13 In a ring: (i) $x(y-z)=xy-xz$ and $(y-z)x=yx-zx$; (ii) $x\cdot 0=0\cdot x=0$; (iii) $x(-y)=(-x)y=-xy$.

Proof. (i) $x(y-z)+xz=x\big((y-z)+z\big)=xy$ by distributivity; adding $-xz$ to both sides gives $x(y-z)=xy-xz$. The other one is symmetric.

(ii) By (i) with $z=y$: $x\cdot 0=x(y-y)=xy-xy=0$. Similarly $0\cdot x=0$.

(iii) $x(-y)+xy=x(-y+y)=x\cdot 0=0$ by (ii), so $x(-y)$ is the symmetric of xy: $x(-y)=-xy$. Similarly $(-x)y=-xy$. ∎

Examples: $\mathbb{Z}$ is a unitary commutative ring, not a field; $\mathbb{Q},\mathbb{R},\mathbb{C}$ are fields; $\mathbb{Z}_n$ is a commutative unitary ring, a field iff n is prime (Seminar 3, Ex. 9); $K[X]$, $M_n(R)$ ($n\geq 2$, not commutative), R^M (Ex. 2), $(\mathcal{P}(M),\Delta,\cap)$ (Boolean ring).

Subgroups

Definition. $H\subseteq G$ is a *subgroup* of the group $(G,\cdot)$, written $H\leq G$, if H is a stable subset and $(H,\cdot)$ is a group.

Theorem (characterization of subgroups). C02 · proof: book Thm 1.6.2, 1.6.3 Let $(G,\cdot)$ be a group and $H\subseteq G$. The following are equivalent:
(a) $H\leq G$;
(b) $H\neq\emptyset$ (equivalently $1\in H$), $\forall x,y\in H: xy\in H$, $\forall x\in H: x^{-1}\in H$;
(c) $H\neq\emptyset$ (equivalently $1\in H$), $\forall x,y\in H: xy^{-1}\in H$.
Additively: $x+y\in H$, $-x\in H$; or $x-y\in H$.

Proof. (a)$\Longrightarrow$(b). $(H,\cdot)$ is a group, so it has an identity $1'$; hence $H\neq\emptyset$. $1'=1$: in G, $1'\cdot 1'=1'=1'\cdot 1$; cancel $1'$ on the left (cancellation in G) $\Longrightarrow 1'=1$, so $1\in H$. Closure: H is stable. *Inverses:* let $x\in H$ and x' its inverse in H: $xx'=x'x=1'=1$. Then x' is also an inverse of x in G, and the inverse in G is unique $\Longrightarrow x'=x^{-1}\in H$.

(b)$\Longrightarrow$(c). For $x,y\in H$: $y^{-1}\in H$, then $xy^{-1}\in H$ by closure.

(c)$\Longrightarrow$(b). Take $x\in H$ (exists since $H\neq\emptyset$). Then $1=xx^{-1}\in H$. For $y\in H$: $y^{-1}=1\cdot y^{-1}\in H$. For $x,y\in H$: $y^{-1}\in H\Longrightarrow xy=x(y^{-1})^{-1}\in H$.

(b)$\Longrightarrow$(a). H is stable. Associativity holds in H because it holds in G. $1\in H$ (as above) is an identity for H. Every $x\in H$ has $x^{-1}\in H$ with $xx^{-1}=x^{-1}x=1$. So $(H,\cdot)$ is a group. ∎

Key idea. A subgroup has the **same** identity and the **same** inverses as the big group (uniqueness of inverses + cancellation). In practice always use (c): two checks.

What to check — "$H\leq G$". (1) $H\subseteq G$ and $1\in H$; (2) take arbitrary $x,y\in H$, write what that means (e.g. $|x|=|y|=1$), and show xy^{-1} satisfies the same condition. To show "not a subgroup": $1\notin H$, or a concrete x,y with $xy^{-1}\notin H$.

Examples: $n\mathbb{Z} \leq (\mathbb{Z}, +)$; $\mathbb{Z} \leq \mathbb{Q} \leq \mathbb{R} \leq \mathbb{C}$ (with $+$); $\mathrm{SL}_n(\mathbb{R}) \leq \mathrm{GL}_n(\mathbb{R})$; $\{z : |z| = 1\} \leq (\mathbb{C}^*, \cdot)$; $U_n \leq (\mathbb{C}^*, \cdot)$.

Subrings and subfields

Definition. $A \subseteq R$ is a *subring* of the ring R if A is stable under $+$ and $\cdot$ and $(A, +, \cdot)$ is a ring. $A \subseteq K$ is a *subfield* of the field K if A is stable and $(A, +, \cdot)$ is a field.

Theorem (characterization of subrings). C02 · proof: book Thm 1.6.8 A is a subring of $(R, +, \cdot) \iff A \neq \emptyset$ $(0 \in A)$, $\forall x, y \in A : x - y \in A$ and $xy \in A$.

Proof. "$\Longrightarrow$" $(A, +)$ is a group, stable in $(R, +)$, hence a subgroup of $(R, +)$; by the characterization of subgroups (additive form (c)), $0 \in A$ and $x - y \in A$. Stability under "$\cdot$" gives $xy \in A$.

"$\Longleftarrow$" The first two conditions say (characterization (c)) that $A \leq (R, +)$, so $(A, +)$ is a group, abelian because R is. The third says A is stable under "$\cdot$". Associativity of "$\cdot$" and the distributive laws hold in A because they hold for all elements of R. Hence $(A, +, \cdot)$ is a ring. ∎

Theorem (characterization of subfields). C02 · proof: book Thm 1.6.9 A is a subfield of the field $K \iff |A| \geq 2$ $(0, 1 \in A)$, $\forall x, y \in A : x - y \in A$, $\forall x, y \in A,\ y \neq 0 : xy^{-1} \in A$.

Proof. "$\Longrightarrow$" $(A, +)$ is a subgroup of $(K, +)$: $0 \in A$ and $x - y \in A$. $(A^*, \cdot)$ is a group, stable in $(K^*, \cdot)$, hence a subgroup: $1 \in A$ (so $|A| \geq 2$) and $y^{-1} \in A$ for $y \in A^*$; then $xy^{-1} \in A$ by stability.

"$\Longleftarrow$" As before, $(A, +) \leq (K, +)$. Stability of "$\cdot$": for $x, y \in A$, if $y = 0$ then $xy = 0 \in A$; if $y \neq 0$, then $y^{-1} = 1 \cdot y^{-1} \in A$ and $y^{-1} \neq 0$, so $xy = x(y^{-1})^{-1} \in A$. The condition $xy^{-1} \in A$ with $1 \in A$ also shows that $(A^*, \cdot)$ is a subgroup of $(K^*, \cdot)$. Associativity, commutativity and distributivity are inherited from K. So A is a field. ∎

Homomorphisms

Definition. $f : G \to G'$ between groups is a *group homomorphism* if $f(xy) = f(x)f(y)$, $\forall x, y \in G$; an *isomorphism* if moreover it is bijective ($G \simeq G'$). $f : R \to R'$ between rings is a *ring homomorphism* if $f(x + y) = f(x) + f(y)$ and $f(xy) = f(x)f(y)$; *unitary* if $f(1) = 1'$.

What to check — "f is an isomorphism". (0) f is well defined and takes values in G' (important for classes such as $\widehat{k}$!); (1) $f(xy) = f(x)f(y)$; (2) injective (for groups: $f(x) = 1' \Rightarrow x = 1$ is enough); (3) surjective. For a ring homomorphism check both $+$ and $\cdot$.

Theorem. C02 · proof: book Thm 1.7.3 If $f : G \to G'$ is a group homomorphism, then (i) $f(1) = 1'$; (ii) $f(x^{-1}) = f(x)^{-1}$, $\forall x \in G$.

Proof. (i) $f(1)f(1) = f(1 \cdot 1) = f(1) = f(1) \cdot 1'$. Cancel $f(1)$ on the left in $G' \Longrightarrow f(1) = 1'$.

(ii) $f(x)f(x^{-1}) = f(xx^{-1}) = f(1) = 1'$ and $f(x^{-1})f(x) = f(x^{-1}x) = 1'$. So $f(x^{-1})$ is an inverse of $f(x)$; by uniqueness, $f(x^{-1}) = f(x)^{-1}$. ∎

For a ring homomorphism, the first condition makes f a group homomorphism of $(R,+)$, so $f(0)=0'$ and $f(-x)=-f(x)$. But $f(1)=1'$ need **not** hold (e.g. the trivial homomorphism $f\equiv 0$).

> **Theorem (unitary ring homomorphisms).** C02 · proof: book Thm 1.7.8 Let $f:R\to R'$ be a ring homomorphism between unitary rings. (i) If f is surjective, then f is unitary. (ii) If f is an isomorphism, then f is unitary. (iii) If f is unitary and x is invertible, then $f(x)$ is invertible and $f(x)^{-1}=f(x^{-1})$.

> ***Proof.*** (i) Let $x'\in R'$; by surjectivity $x'=f(x)$ for some $x\in R$. Then $x'f(1)=f(x)f(1)=f(x\cdot 1)=f(x)=x'$ and similarly $f(1)x'=x'$. So $f(1)$ is an identity of R'; by uniqueness, $f(1)=1'$.
>
> (ii) An isomorphism is surjective; apply (i).
>
> (iii) $f(x)f(x^{-1})=f(xx^{-1})=f(1)=1'$ and $f(x^{-1})f(x)=f(1)=1'$. ∎

Determinants — facts used in exercises

The determinant over a field K is defined as for real matrices, $\det A=\sum_{\sigma\in S_n}\operatorname{sgn}(\sigma)a_{1\sigma(1)}\cdots a_{n\sigma(n)}$. The course uses the following properties (book Thm 1.8.6–1.8.16, stated without proof, as for real matrices):

- Laplace expansion along any row or column; $\det A=\det A^T$; $\det(AB)=\det A\cdot\det B$; $\det(\alpha A)=\alpha^n\det A$.
- Two equal rows $\Longrightarrow\det=0$; swapping two rows changes the sign; adding a multiple of a row to another row does not change $\det$; triangular matrix: product of the diagonal.
- A is invertible $\iff\det A\neq 0$, and then $A^{-1}=(\det A)^{-1}A^*$, $\det(A^{-1})=(\det A)^{-1}$.
- $\operatorname{rank}A$ = maximal order of a non-zero minor; for $m=n$, A invertible $\iff\operatorname{rank}A=n$.

Chapter 3 — Solutions (Seminar 3)

Ex. 1

Operation: if $f,g\in S_M$, then $f\circ g:M\to M$ is bijective, because it has the inverse $g^{-1}\circ f^{-1}$:

$$(f\circ g)\circ(g^{-1}\circ f^{-1})=f\circ(g\circ g^{-1})\circ f^{-1}=f\circ f^{-1}=1_M,\quad (g^{-1}\circ f^{-1})\circ(f\circ g)=1_M.$$

Associative: composition of functions is associative: $((f\circ g)\circ h)(x)=f(g(h(x)))=(f\circ(g\circ h))(x)$.

Identity: $1_M\in S_M$ and $f\circ 1_M=1_M\circ f=f$.

Inverses: f bijective $\Longrightarrow f^{-1}$ exists, is bijective (its inverse is f) and $f\circ f^{-1}=f^{-1}\circ f=1_M$.

$\Longrightarrow(S_M,\circ)$ **is a group.**

It is not abelian if $|M|\geq 3$: for $M\supseteq\{1,2,3\}$, the transpositions $f=(1\,2)$, $g=(2\,3)$ (fixing all other points) give $(f\circ g)(1)=2$ but $(g\circ f)(1)=3$.

Ex. 2

All operations are defined *pointwise*, so every law holding in R holds in R^M at each point x.

$(R^M, +)$ **abelian group.** Associative and commutative: $((f+g)+h)(x) = (f(x)+g(x))+h(x) = f(x)+(g(x)+h(x)) = (f+(g+h))(x)$, and $(f+g)(x) = g(x)+f(x)$. Zero: $\theta(x) = 0$, since $(f+\theta)(x) = f(x)$. Symmetric: $(-f)(x) = -f(x)$, since $(f+(-f))(x) = 0 = \theta(x)$.

$(R^M, \cdot)$ **semigroup.** $((fg)h)(x) = (f(x)g(x))h(x) = f(x)(g(x)h(x)) = (f(gh))(x)$.

Distributivity. $(f(g+h))(x) = f(x)\big(g(x)+h(x)\big) = f(x)g(x)+f(x)h(x) = (fg+fh)(x)$; the right law is analogous.

$\Longrightarrow$ $(R^M, +, \cdot)$ **is a ring.**

Commutative: if R is commutative, $(fg)(x) = f(x)g(x) = g(x)f(x) = (gf)(x) \Longrightarrow$ yes. **Identity:** if R has 1, the constant function $u(x) = 1$ satisfies $(fu)(x) = f(x)\cdot 1 = f(x) \Longrightarrow$ yes, u is the identity. (Conversely, since $M \neq \emptyset$, these properties of R^M force the same properties for R, using constant functions.)

Ex. 3

$H \subseteq \mathbb{C}^*$, because $|z| = 1 \Longrightarrow z \neq 0$. Use the characterization (c):

- $1 \in H$ since $|1| = 1$;
- $z, w \in H \Longrightarrow |zw^{-1}| = \dfrac{|z|}{|w|} = \dfrac{1}{1} = 1 \Longrightarrow zw^{-1} \in H$.

$\Longrightarrow$ $H \leq (\mathbb{C}^*, \cdot)$ **(the circle group).**

In $(\mathbb{C}, +)$ the identity is 0, and $|0| = 0 \neq 1$, so $0 \notin H$ (also: $1 \in H$ but $1+1 = 2 \notin H$).

$\Longrightarrow$ H **is not a subgroup of** $(\mathbb{C}, +)$**.**

Ex. 4

$U_n \subseteq \mathbb{C}^*$: if $z^n = 1$ then $z \neq 0$. Characterization (c):

- $1^n = 1 \Longrightarrow 1 \in U_n$;
- $z, w \in U_n \Longrightarrow (zw^{-1})^n = z^n(w^n)^{-1} = 1 \cdot 1^{-1} = 1$ (powers of commuting elements) $\Longrightarrow zw^{-1} \in U_n$.

$\Longrightarrow$ $U_n \leq (\mathbb{C}^*, \cdot)$**.**

Explicitly $U_n = \{\varepsilon_k \mid k = 0, \ldots, n-1\}$, $\varepsilon_k = \cos\frac{2k\pi}{n} + i\sin\frac{2k\pi}{n} = \varepsilon_1^k$ (de Moivre), $|U_n| = n$.

Ex. 5

(i) Let $A, B \in \mathrm{GL}_n(\mathbb{C})$. Then $\det(AB) = \det A \cdot \det B \neq 0$ (product of non-zero complex numbers) $\Longrightarrow AB \in \mathrm{GL}_n(\mathbb{C})$: stable.

(ii) Matrix multiplication is associative. $I_n \in \mathrm{GL}_n(\mathbb{C})$ ($\det I_n = 1$) is the identity. For $A \in \mathrm{GL}_n(\mathbb{C})$, $\det A \neq 0$, so A is invertible with $A^{-1} = (\det A)^{-1}A^*$, and $\det(A^{-1}) = (\det A)^{-1} \neq 0 \Longrightarrow A^{-1} \in \mathrm{GL}_n(\mathbb{C})$.

$\Longrightarrow$ $(\mathrm{GL}_n(\mathbb{C}), \cdot)$ **is a group (not abelian for** $n \geq 2$**).**

(iii) $\mathrm{SL}_n(\mathbb{C}) \subseteq \mathrm{GL}_n(\mathbb{C})$. $\det I_n = 1 \Longrightarrow I_n \in \mathrm{SL}_n(\mathbb{C})$. For $A, B \in \mathrm{SL}_n(\mathbb{C})$:

$$\det(AB^{-1}) = \det A \cdot (\det B)^{-1} = 1 \cdot 1^{-1} = 1.$$

$\Longrightarrow$ $\mathrm{SL}_n(\mathbb{C}) \leq \mathrm{GL}_n(\mathbb{C})$**.**

Ex. 6

Characterization of subrings: $0 \in A$, $x - y \in A$, $xy \in A$.

(i) $0 = 0 + 0i \in \mathbb{Z}[i]$. For $a + bi$, $c + di \in \mathbb{Z}[i]$:

$$(a + bi) - (c + di) = (a - c) + (b - d)i, \qquad (a + bi)(c + di) = (ac - bd) + (ad + bc)i,$$

with $a - c$, $b - d$, $ac - bd$, $ad + bc \in \mathbb{Z}$.

$\Longrightarrow$ $\mathbb{Z}[i]$ is a subring of $\mathbb{C}$ (not a subfield: $2 \in \mathbb{Z}[i]$ but $\frac{1}{2} \notin \mathbb{Z}[i]$).

(ii) $O_2 \in M$ ($a = b = c = 0$). For two elements of M:

$$\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} - \begin{pmatrix} a' & b' \\ 0 & c' \end{pmatrix} = \begin{pmatrix} a - a' & b - b' \\ 0 & c - c' \end{pmatrix}, \qquad \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \begin{pmatrix} a' & b' \\ 0 & c' \end{pmatrix} = \begin{pmatrix} aa' & ab' + bc' \\ 0 & cc' \end{pmatrix}.$$

Both are upper triangular.

$\Longrightarrow$ M is a subring of $M_2(\mathbb{R})$.

Ex. 7

(i) $z \neq 0 \Longrightarrow |z| \neq 0$, so $f(z) \in \mathbb{R}^*$. For $z, w \in \mathbb{C}^*$: $f(zw) = |zw| = |z|\,|w| = f(z)f(w)$.

$\Longrightarrow$ f is a group homomorphism (not injective: $f(1) = f(-1)$; image $(0, \infty)$).

(ii) *Values in* $\mathrm{GL}_2(\mathbb{R})$*:* $\det g(a + bi) = a^2 + b^2 \neq 0$ for $a + bi \neq 0$.

Homomorphism: for $z = a + bi$, $w = c + di$, $zw = (ac - bd) + (ad + bc)i$ and

$$g(z)g(w) = \begin{pmatrix} a & b \\ -b & a \end{pmatrix} \begin{pmatrix} c & d \\ -d & c \end{pmatrix} = \begin{pmatrix} ac - bd & ad + bc \\ -(ad + bc) & ac - bd \end{pmatrix} = g(zw).$$

$\Longrightarrow$ g is a group homomorphism (even injective).

Ex. 8

Let $\varepsilon = \cos\frac{2\pi}{n} + i\sin\frac{2\pi}{n}$, so $U_n = \{\varepsilon^0, \varepsilon^1, \ldots, \varepsilon^{n-1}\}$ and $\varepsilon^k = 1 \iff n \mid k$. Define $f : \mathbb{Z}_n \to U_n$, $f(\widehat{k}) = \varepsilon^k$.

Well defined (independent of the representative): $\widehat{k} = \hat{l} \Longrightarrow k = l + nq \Longrightarrow \varepsilon^k = \varepsilon^l(\varepsilon^n)^q = \varepsilon^l$. Also $(\varepsilon^k)^n = (\varepsilon^n)^k = 1$, so $f(\widehat{k}) \in U_n$.

Homomorphism from $(\mathbb{Z}_n, +)$ to $(U_n, \cdot)$: $f(\widehat{k} + \hat{l}) = f(\widehat{k + l}) = \varepsilon^{k+l} = \varepsilon^k\varepsilon^l = f(\widehat{k})f(\hat{l})$.

Injective: $f(\widehat{k}) = f(\hat{l}) \Longrightarrow \varepsilon^{k-l} = 1 \Longrightarrow \cos\frac{2\pi(k-l)}{n} = 1 \Longrightarrow n \mid k - l \Longrightarrow \widehat{k} = \hat{l}$.

Surjective: every element of U_n is some $\varepsilon^k = f(\widehat{k})$. (Alternatively: injective between two sets with n elements.)

$\Longrightarrow$ f is a group isomorphism, $(\mathbb{Z}_n, +) \simeq (U_n, \cdot)$.

Ex. 9

(i) "$\Longleftarrow$" If $\gcd(a, n) = 1$, Bézout gives $b, q \in \mathbb{Z}$ with $ab + nq = 1$. Then $\widehat{a}\,\widehat{b} = \widehat{ab} = \widehat{1 - nq} = \widehat{1}$ (and $\widehat{b}\widehat{a} = \widehat{1}$, $\mathbb{Z}_n$ being commutative) $\Longrightarrow \widehat{a}$ is invertible.

"$\Longrightarrow$" If $\widehat{a}\widehat{b} = \widehat{1}$, then $n \mid ab - 1$, i.e. $ab - 1 = nq$ for some $q \in \mathbb{Z}$. Any common divisor d of a and n divides $ab - nq = 1 \Longrightarrow d = \pm 1 \Longrightarrow \gcd(a, n) = 1$.

(ii) $(\mathbb{Z}_n, +, \cdot)$ is a commutative unitary ring with $\widehat{1} \neq \widehat{0}$ ($n \geq 2$). So it is a field $\iff$ every $\widehat{a} \neq \widehat{0}$, i.e. $a \in \{1, \dots, n-1\}$, is invertible $\iff$ (by (i)) $\gcd(a, n) = 1$ for all $a \in \{1, \dots, n-1\}$ $\iff$ n is prime.

(Last equivalence: if n is prime, its only positive divisors are $1, n$, so $\gcd(a, n) = 1$ for $1 \leq a \leq n-1$. If n is not prime, $n = ab$ with $1 < a < n$, and $\gcd(a, n) = a \neq 1$ — indeed $\widehat{a}\widehat{b} = \widehat{0}$ with $\widehat{a}, \widehat{b} \neq \widehat{0}$.)

$\Longrightarrow$ $\mathbb{Z}_n$ is a field $\iff$ n is prime.

Ex. 10

M is a subring of $M_2(\mathbb{R})$: $O_2 \in M$ ($a = b = 0$); the difference of two matrices of M is in M (subtract entries); the product

$$\begin{pmatrix} a & b \\ -b & a \end{pmatrix} \begin{pmatrix} c & d \\ -d & c \end{pmatrix} = \begin{pmatrix} ac - bd & ad + bc \\ -(ad + bc) & ac - bd \end{pmatrix} \in M. \qquad (*)$$

Field: $(*)$ is symmetric in $(a, b) \leftrightarrow (c, d)$ $\Longrightarrow$ commutative. $I_2 \in M$ ($a = 1, b = 0$) is the identity, $I_2 \neq O_2$. If $A \neq O_2$, then $\det A = a^2 + b^2 > 0$ and

$$A^{-1} = \frac{1}{a^2 + b^2} \begin{pmatrix} a & -b \\ b & a \end{pmatrix} \in M \quad (\text{entries } a', b' \text{ with } a' = \tfrac{a}{a^2+b^2},\ b' = \tfrac{-b}{a^2+b^2}).$$

$\Longrightarrow$ $(M, +, \cdot)$ is a field.

Isomorphism: $\varphi : \mathbb{C} \to M$, $\varphi(a + bi) = \begin{pmatrix} a & b \\ -b & a \end{pmatrix}$.

- $\varphi(z + w) = \varphi(z) + \varphi(w)$: entries add;
- $\varphi(zw) = \varphi(z)\varphi(w)$: this is $(*)$, since $(a + bi)(c + di) = (ac - bd) + (ad + bc)i$;
- bijective: a matrix of M determines a, b uniquely (first row), so each matrix comes from exactly one $a + bi$.

$\Longrightarrow$ $(M, +, \cdot) \simeq (\mathbb{C}, +, \cdot)$.

Results at a glance — Seminar 3

Ex.	Answer / key step
1	$(f \circ g)^{-1} = g^{-1} \circ f^{-1}$; identity 1_M
2	pointwise; R^M commutative / unitary when R is
3, 4, 5(iii)	$1 \in H$ and $xy^{-1} \in H$
5(i)	$\det(AB) = \det A \det B$
6	$0 \in A$, $x - y \in A$, $xy \in A$
8	$f(\widehat{k}) = \varepsilon^k$, well defined since $\varepsilon^n = 1$
9	Bézout; $\mathbb{Z}_n$ field $\Longleftrightarrow$ n prime
10	$\varphi(a + bi) = \begin{pmatrix} a & b \\ -b & a \end{pmatrix}$

Chapter 4 — Vector spaces and subspaces: theory

Course 3 · book §2.1–2.2. From now on K is a field.

Vector spaces

Definition (vector space). A vector space over K (K-vector space) is an abelian group $(V,+)$ together with an external operation $\cdot : K \times V \to V$, $(k,v) \mapsto kv$, such that for all $k, k_1, k_2 \in K$ and $v, v_1, v_2 \in V$:

(L1) $k(v_1+v_2) = kv_1 + kv_2$; (L2) $(k_1+k_2)v = k_1v + k_2v$; (L3) $(k_1k_2)v = k_1(k_2v)$; (L4) $1 \cdot v = v$.

Elements of K are *scalars*, elements of V are *vectors*. Notation: ${}_KV$ or $(V, K, +, \cdot)$.

What to check — "V is a K-vector space".

1. "$+$" is an operation on V and $kv \in V$ for all $k \in K$, $v \in V$;
2. $(V,+)$ is an abelian group: associative, commutative, a zero vector (find it — it may not be "0", see Ex. 4), symmetric of each vector;
3. (L1)–(L4) for arbitrary scalars and vectors.

For operations defined componentwise / coefficientwise / pointwise, each axiom reduces to the same property in the field K. To **disprove**, give concrete numbers violating one axiom.

Standard examples: K^n (canonical vector space), $K[X]$, $K_n[X]$, $M_{m,n}(K)$, $K^A = \{f : A \to K\}$, V_2, V_3 (geometric vectors), $V \times V'$ (direct product), K over a subfield ($\mathbb{R}$ over $\mathbb{Q}$, $\mathbb{C}$ over $\mathbb{R}$), the zero space $\{0\}$. A finite group $(\mathbb{Z}_n, +)$, $n \geq 2$, admits **no** real vector space structure: for $v \neq 0$, $a \mapsto av$ would be an injective map $\mathbb{R} \to \mathbb{Z}_n$ (C03).

Theorem (computation rules). C03 · proof: (i),(ii) course, (iii),(iv) homework For all $k, k' \in K$, $v, v' \in V$:
(i) $k \cdot 0 = 0 \cdot v = 0$; (ii) $k(-v) = (-k)v = -kv$; (iii) $k(v-v') = kv - kv'$; (iv) $(k-k')v = kv - k'v$.

Proof. Throughout we use: in the group $(V,+)$, if $a+b=b$ then $a=0$ (add $-b$), and if $a+b=0$ then $a=-b$.

(i) $k\cdot 0 + kv \overset{(L1)}{=} k(0+v) = kv \implies k\cdot 0 = 0$. Also $0\cdot v + kv \overset{(L2)}{=} (0+k)v = kv \implies 0\cdot v = 0$.

(ii) $kv + k(-v) \overset{(L1)}{=} k(v+(-v)) = k\cdot 0 \overset{(i)}{=} 0 \implies k(-v) = -kv$. And $kv + (-k)v \overset{(L2)}{=} (k+(-k))v = 0 \cdot v = 0 \implies (-k)v = -kv$.

(iii) $k(v-v') + kv' \overset{(L1)}{=} k\big((v-v')+v'\big) = kv \implies k(v-v') = kv - kv'$.

(iv) $(k-k')v + k'v \overset{(L2)}{=} \big((k-k')+k'\big)v = kv \implies (k-k')v = kv - k'v$. ∎

Theorem. C03 · proof: course $kv = 0 \iff k = 0$ or $v = 0$.

Proof. "$\impliedby$" is (i) above. "$\implies$" Let $kv = 0$ and suppose $k \neq 0$. Then k^{-1} exists in the field K and

$$v \overset{(L4)}{=} 1 \cdot v = (k^{-1}k)v \overset{(L3)}{=} k^{-1}(kv) = k^{-1} \cdot 0 \overset{(i)}{=} 0. \quad \blacksquare$$

Key idea. Every rule is proved the same way: apply (L1) or (L2) to "something + something", get an equation $a + b = b$ in the group $(V, +)$, cancel. The second theorem is where we need K to be a **field** (to invert k).

Subspaces

Definition. $S \subseteq V$ is a *subspace* of V ($S \leq V$) if (i) $S \neq \emptyset$; (ii) $v_1 + v_2 \in S$ for all $v_1, v_2 \in S$; (iii) $kv \in S$ for all $k \in K$, $v \in S$. Every subspace is a subgroup of $(V, +)$, so $0 \in S$.

Theorem (characterization of subspaces). C03 · proof: course

$$S \leq V \iff \begin{cases} S \neq \emptyset \ \text{(equivalently } 0 \in S) \\ \forall k_1, k_2 \in K,\ \forall v_1, v_2 \in S:\ k_1 v_1 + k_2 v_2 \in S. \end{cases}$$

Proof. "$\Longrightarrow$" Take $v_1 \in S$ (exists, $S \neq \emptyset$) and $k = 0$: by (iii), $0 = 0 \cdot v_1 \in S$. For $k_1, k_2 \in K$, $v_1, v_2 \in S$: $k_1 v_1, k_2 v_2 \in S$ by (iii), so $k_1 v_1 + k_2 v_2 \in S$ by (ii).

"$\Longleftarrow$" $S \neq \emptyset$. Take $k_1 = k_2 = 1$: $v_1 + v_2 = 1v_1 + 1v_2 \in S$ (using (L4)), which is (ii). Take $k_2 = 0$: $k_1 v_1 = k_1 v_1 + 0 \cdot v_2 \in S$, which is (iii). ∎

What to check — "$S \leq V$". (1) $0 \in S$ (if $0 \notin S$, stop: not a subspace); (2) take arbitrary $k_1, k_2 \in K$ and $v_1, v_2 \in S$, write the condition defining S for v_1, v_2, and verify it for $k_1 v_1 + k_2 v_2$ — usually it is a **linear homogeneous** condition, and the verification is "$= k_1 \cdot 0 + k_2 \cdot 0 = 0$".
Typical "no" answers: a non-homogeneous equation ($= 1$: $0 \notin S$), an "or" condition (union of two subspaces, not closed under $+$), a bounded set or integrality condition (not closed under scalars), $\deg f = n$ ($0 \notin S$).

Geometric picture. The subspaces of $\mathbb{R}^3$ are exactly $\{0\}$, the lines through the origin, the planes through the origin and $\mathbb{R}^3$. A plane or line *not* through the origin is never a subspace.

Chapter 4 — Solutions (Seminar 4)

Ex. 1

Write two polynomials with the same number of coefficients (add zero coefficients if needed): $f = \sum_{i=0}^{n} a_i X^i$, $g = \sum_{i=0}^{n} b_i X^i$. Then $f + g = \sum (a_i + b_i) X^i$ and $kf = \sum (k a_i) X^i$.

$(K[X], +)$ **is an abelian group** (it is the additive group of the ring $K[X]$): coefficientwise addition is associative and commutative because $+$ in K is; the zero is the zero polynomial; $-f = \sum (-a_i) X^i$.

Axioms — each reduces to a property of the field K in every coefficient:

- (L1) $k(f+g) = \sum k(a_i+b_i)X^i = \sum(ka_i+kb_i)X^i = kf+kg$ (distributivity in K);
- (L2) $(k_1+k_2)f = \sum(k_1+k_2)a_iX^i = \sum(k_1a_i+k_2a_i)X^i = k_1f+k_2f$;
- (L3) $(k_1k_2)f = \sum(k_1k_2)a_iX^i = \sum k_1(k_2a_i)X^i = k_1(k_2f)$ (associativity in K);
- (L4) $1\cdot f = \sum(1\cdot a_i)X^i = f$.

$\Longrightarrow$ $K[X]$ **is a** K**-vector space.**

Ex. 2

Identical to Ex. 1, but entrywise: for $A=(a_{ij})$, $B=(b_{ij})$, $A+B=(a_{ij}+b_{ij})$, $kA=(ka_{ij})$.

$(M_{m,n}(K),+)$ is an abelian group (zero matrix O, $-A=(-a_{ij})$). (L1): $k(A+B)=(k(a_{ij}+b_{ij}))=(ka_{ij}+kb_{ij})=kA+kB$; (L2): $((k_1+k_2)a_{ij})=(k_1a_{ij}+k_2a_{ij})$; (L3): $((k_1k_2)a_{ij})=(k_1(k_2a_{ij}))$; (L4): $(1a_{ij})=(a_{ij})$.

$\Longrightarrow$ $M_{m,n}(K)$ **is a** K**-vector space.**

Ex. 3

Pointwise, at each $x\in A$:

- $(K^A,+)$ abelian group: associativity and commutativity hold at every point; zero = the function $\theta(x)=0$; symmetric: $(-f)(x)=-f(x)$.
- (L1) $(k(f+g))(x)=k(f(x)+g(x))=kf(x)+kg(x)=(kf+kg)(x)$;
- (L2) $((k_1+k_2)f)(x)=(k_1+k_2)f(x)=k_1f(x)+k_2f(x)=(k_1f+k_2f)(x)$;
- (L3) $((k_1k_2)f)(x)=(k_1k_2)f(x)=k_1(k_2f(x))=(k_1(k_2f))(x)$;
- (L4) $(1f)(x)=f(x)$.

Two functions are equal iff they are equal at every point, so each identity holds in K^A.

$\Longrightarrow$ K^A **is a** K**-vector space.**

Ex. 4

$V=(0,\infty)$, "addition" $x\perp y=xy$, "scalar multiplication" $k\top x=x^k$.

Operations: $x,y>0\Rightarrow xy>0$; $x>0\Rightarrow x^k>0$ for every real k.

$(V,\perp)$ **abelian group:** multiplication of reals is associative and commutative; the **zero vector is** 1 ($x\perp 1=x$); the symmetric of x is $\frac{1}{x}\in V$ ($x\perp\frac{1}{x}=1$).

Axioms (power rules for positive reals):

- (L1) $k\top(x\perp y)=(xy)^k=x^ky^k=(k\top x)\perp(k\top y)$;
- (L2) $(k_1+k_2)\top x=x^{k_1+k_2}=x^{k_1}x^{k_2}=(k_1\top x)\perp(k_2\top x)$;
- (L3) $(k_1k_2)\top x=x^{k_1k_2}=(x^{k_2})^{k_1}=k_1\top(k_2\top x)$;
- (L4) $1\top x=x^1=x$.

$\Longrightarrow$ $(V,\perp,\top)$ **is a real vector space.**

Remark: $\ln: V\to\mathbb{R}$ turns $\perp$ into $+$ and $\top$ into the usual scalar multiplication ($\ln(xy)=\ln x+\ln y$, $\ln x^k=k\ln x$); it is an isomorphism $V\simeq\mathbb{R}$.

Ex. 5

(i) The addition is not commutative:

$$(0,1)+(0,0)=(0,\,1+0)=(0,1), \qquad (0,0)+(0,1)=(0,\,0+2)=(0,2),$$

and $(0,1)\neq(0,2)$ because $1\neq 2$ in any field ($1=2 \iff 1=0$). So $(V,+)$ is not an abelian group.

$\Longrightarrow V$ is not a vector space (in no field K).

Associativity fails too: $((0,0)+(0,0))+(0,1)=(0,2)$ but $(0,0)+((0,0)+(0,1))=(0,4)$ — different unless $2=0$.

(ii) Here $(V,+)$ is the usual abelian group, (L1), (L3), (L4) hold, but (L2) fails:

$$(1+1)\cdot(0,1)=(0,1), \qquad 1\cdot(0,1)+1\cdot(0,1)=(0,1)+(0,1)=(0,2)\neq(0,1).$$

$\Longrightarrow V$ is not a vector space.

Ex. 6

(i) Let $x\in V$. By (L4) and (L2) (applied repeatedly),

$$\underbrace{x+\cdots+x}_{p}=\underbrace{\hat{1}x+\cdots+\hat{1}x}_{p}=(\underbrace{\hat{1}+\cdots+\hat{1}}_{p})x=\hat{p}\,x=\hat{0}\,x=0.$$

(ii) No. If $(\mathbb{Z},+)$ were a vector space over $\mathbb{Z}_p$, then by (i), with $x=1$: $\underbrace{1+\cdots+1}_{p}=p=0$ in $\mathbb{Z}$, which is false.

$\Longrightarrow$ there is no such scalar multiplication.

Ex. 7

(i) A: yes. $0\in A$. If $v_1=(0,y_1,z_1)$, $v_2=(0,y_2,z_2)\in A$, then $k_1v_1+k_2v_2=(0,\;k_1y_1+k_2y_2,\;k_1z_1+k_2z_2)\in A$.

(ii) B: no. $(1,0,0)\in B$ ($z=0$) and $(0,0,1)\in B$ ($x=0$), but $(1,0,0)+(0,0,1)=(1,0,1)\notin B$. ($B$ is the union of two planes.)

(iii) C: no. $(1,0,0)\in C$ but $\frac{1}{2}(1,0,0)=(\frac{1}{2},0,0)\notin C$.

(iv) D: yes. $0\in D$. If $x_i+y_i+z_i=0$ ($i=1,2$), then $(k_1x_1+k_2x_2)+(k_1y_1+k_2y_2)+(k_1z_1+k_2z_2)=k_1\cdot 0+k_2\cdot 0=0$.

(v) E: no. $0+0+0=0\neq 1\Longrightarrow 0\notin E$.

(vi) F: yes. $0\in F$. If $v_i=(t_i,t_i,t_i)$, then $k_1v_1+k_2v_2=(t,t,t)$ with $t=k_1t_1+k_2t_2$.

$\Longrightarrow$ Subspaces: A, D, F (two planes and a line through the origin).

Ex. 8

(i) No. $1\in[-1,1]$, but $2\cdot 1=2\notin[-1,1]$.

(ii) No. $(1,0)$ lies in the disk, $2\cdot(1,0)=(2,0)$ does not ($4>1$).

(iii) Over $\mathbb{Q}$: **yes** — O_2 is in the set, and $k_1\begin{pmatrix}a_1 & b_1\\ 0 & c_1\end{pmatrix}+k_2\begin{pmatrix}a_2 & b_2\\ 0 & c_2\end{pmatrix}=\begin{pmatrix}k_1a_1+k_2a_2 & k_1b_1+k_2b_2\\ 0 & k_1c_1+k_2c_2\end{pmatrix}$ has rational entries when $k_1,k_2\in\mathbb{Q}$. Over $\mathbb{R}$: **no** — I_2 is in the set, but $\sqrt{2}\,I_2$ has irrational

entries.

(iv) Yes. The zero function is continuous, and if f, g are continuous then $k_1 f + k_2 g$ is continuous (sum and constant multiples of continuous functions).

Ex. 9

Convention: $\deg 0 = -\infty$, and $\deg(f+g) \le \max(\deg f, \deg g)$, $\deg(kf) \le \deg f$.

(i) Yes. $0 \in K_n[X]$. If $\deg f, \deg g \le n$, then $\deg(k_1 f + k_2 g) \le \max(\deg f, \deg g) \le n$.

(ii) No. $\deg 0 = -\infty \neq n$, so $0 \notin K'_n[X]$. (Also, for $n \ge 1$: X^n and $-X^n + 1$ have degree n, but their sum 1 has degree 0.)

Ex. 10

Let the system be $a_{11}x + a_{12}y = 0$, $a_{21}x + a_{22}y = 0$, and $S \subseteq \mathbb{R}^2$ its set of solutions.

$(0,0) \in S$. Let $(x_1, y_1), (x_2, y_2) \in S$ and $k_1, k_2 \in \mathbb{R}$. For the vector $k_1(x_1, y_1) + k_2(x_2, y_2) = (k_1 x_1 + k_2 x_2,\ k_1 y_1 + k_2 y_2)$ and each $i = 1, 2$:

$$a_{i1}(k_1 x_1 + k_2 x_2) + a_{i2}(k_1 y_1 + k_2 y_2) = k_1(a_{i1}x_1 + a_{i2}y_1) + k_2(a_{i1}x_2 + a_{i2}y_2) = k_1 \cdot 0 + k_2 \cdot 0 = 0.$$

$\Longrightarrow S \le \mathbb{R}^2$**.**

Faster, after Course 4: $S = \operatorname{Ker} f$ for the linear map $f(x, y) = (a_{11}x + a_{12}y,\ a_{21}x + a_{22}y)$, and a kernel is always a subspace. The same holds for any homogeneous system with m equations and n unknowns.

Results at a glance — Seminar 4

Ex.	Answer
1–3	vector spaces (axioms coefficientwise / entrywise / pointwise)
4	vector space; zero vector 1, symmetric $1/x$
5	(i) no: $+$ not commutative; (ii) no: (L2) fails
6	$p\,x = 0$; no structure on $\mathbb{Z}$
7	subspaces: A, D, F; not: B, C, E
8	(i) no; (ii) no; (iii) yes over $\mathbb{Q}$, no over $\mathbb{R}$; (iv) yes
9	(i) yes; (ii) no
10	yes (kernel of a linear map)

Chapter 5 — Generated subspaces, sums, linear maps: theory

Course 4 · book §2.3–2.4.

Intersection of subspaces

Theorem. C04 · proof: course If $(S_i)_{i\in I}$ is a non-empty family of subspaces of V, then $\bigcap_{i\in I} S_i \leq V$.

Proof. $0 \in S_i$ for every i, so $0 \in \bigcap S_i \neq \emptyset$. Let $k_1, k_2 \in K$ and $x, y \in \bigcap S_i$. Then $x, y \in S_i$ for every i, and since each S_i is a subspace, $k_1x + k_2y \in S_i$ for every i. Hence $k_1x + k_2y \in \bigcap S_i$. By the characterization of subspaces, $\bigcap S_i \leq V$. ∎

Warning. A union of subspaces is in general **not** a subspace: $S = \{(x, 0)\}$, $T = \{(0, y)\}$ in $\mathbb{R}^2$; $(1,0), (0,1) \in S \cup T$ but $(1,1) \notin S \cup T$. The correct "smallest subspace containing both" is $S + T$.

Generated subspace

Definition. For $X \subseteq V$, $\langle X\rangle = \bigcap\{S \leq V \mid X \subseteq S\}$, the subspace *generated* (spanned) by X. It is the smallest subspace containing X; $\langle\emptyset\rangle = \{0\}$; $\langle S\rangle = S$ if $S \leq V$. V is *finitely generated* if $V = \langle v_1, \ldots, v_n\rangle$ for some vectors (a *system of generators*). A *linear combination* of $v_1, \ldots, v_n$ is $k_1v_1 + \cdots + k_nv_n$ with $k_i \in K$.

Theorem (characterization of the generated subspace). C04 · proof: course For $\emptyset \neq X \subseteq V$,

$$\langle X\rangle = \{k_1v_1 + \cdots + k_nv_n \mid n \in \mathbb{N}^*,\ k_i \in K,\ v_i \in X\},$$

the set of all finite linear combinations of vectors of X. In particular $\langle x_1, \ldots, x_n\rangle = \{k_1x_1 + \cdots + k_nx_n \mid k_i \in K\}$.

Proof. Let L be the set on the right. Since $\langle X\rangle$ is the smallest subspace containing X, it is enough to prove: (i) $L \leq V$; (ii) $X \subseteq L$; (iii) $L \subseteq S$ for every subspace $S \supseteq X$. Then $\langle X\rangle \subseteq L$ by (i),(ii) and the definition, and $L \subseteq \langle X\rangle$ by (iii) applied to $S = \langle X\rangle$.

(i) For $v \in X$, $v = 1 \cdot v \in L$, so $L \neq \emptyset$. If $v = \sum_{i=1}^{n} k_iv_i$ and $v' = \sum_{j=1}^{m} k'_jv'_j$ are in L and $k, k' \in K$, then $kv + k'v' = \sum_i (kk_i)v_i + \sum_j (k'k'_j)v'_j$ is again a finite linear combination of vectors of X, so it is in L. Hence $L \leq V$.

(ii) $v = 1 \cdot v \in L$ for every $v \in X$.

(iii) If $S \leq V$ and $X \subseteq S$, then each $v_i \in S$, and S is closed under linear combinations, so $k_1v_1 + \cdots + k_nv_n \in S$. Hence $L \subseteq S$. ∎

Key idea. To prove $\langle Y\rangle \subseteq \langle X\rangle$, it is enough to show $Y \subseteq \langle X\rangle$, i.e. every vector of Y is a linear combination of vectors of X (then $\langle X\rangle$ is a subspace containing Y, and $\langle Y\rangle$ is the smallest one). This is how Seminar 5, Ex. 3 is solved.

Recipe — "write S as a generated subspace". Solve the equations defining S (express some coordinates through the free ones), write the general vector, split it as a sum over free parameters: $S = \{(y + z, y, z)\} = \{y(1,1,0) + z(1,0,1)\} = \langle(1,1,0), (1,0,1)\rangle$.

Sum and direct sum

Definition. For $S, T \leq V$: $S + T = \{s + t \mid s \in S,\ t \in T\}$. If $S \cap T = \{0\}$, the sum is *direct*, written $S \oplus T$.

Theorem. C04 · proof: course $S + T = \langle S \cup T \rangle$. In particular $S + T \leq V$.

Proof. "$\subseteq$" If $v = s + t$, then $v = 1 \cdot s + 1 \cdot t$ is a linear combination of vectors of $S \cup T$, so $v \in \langle S \cup T \rangle$.

"$\supseteq$" Let $v = \sum_{i=1}^{n} k_i v_i$ with $v_i \in S \cup T$. Split the indices: $I = \{i \mid v_i \in S\}$, $J = \{j \mid v_j \in T \setminus S\}$. Then $v = \sum_{i \in I} k_i v_i + \sum_{j \in J} k_j v_j$; the first sum is in S and the second in T (subspaces are closed under linear combinations). So $v \in S + T$. ∎

Theorem (direct sum). C04 · proof: course $V = S \oplus T \iff$ every $v \in V$ can be written uniquely as $v = s + t$ with $s \in S$, $t \in T$.

Proof. "$\Longrightarrow$" Since $V = S + T$, a writing $v = s + t$ exists. If also $v = s' + t'$, then $s - s' = t' - t$; the left side is in S, the right side in T, so both lie in $S \cap T = \{0\}$. Hence $s = s'$, $t = t'$.

"$\Longleftarrow$" Existence of the writings gives $V \subseteq S + T$, and $S + T \subseteq V$ always, so $V = S + T$. If there were $0 \neq v \in S \cap T$, then $v = v + 0$ (with $v \in S$, $0 \in T$) and $v = 0 + v$ (with $0 \in S$, $v \in T$) would be two different writings — contradiction. So $S \cap T = \{0\}$. ∎

What to check — "$V = S \oplus T$". (1) $S, T \leq V$; (2) $S \cap T = \{0\}$: take v in both, show $v = 0$; (3) $V = S + T$: for an arbitrary v, *construct* $s \in S$, $t \in T$ with $v = s + t$ (or, after Course 6: compare dimensions).

Linear maps

Definition. $f : V \to V'$ (V, V' over the same K) is a *K-linear map* if $f(v_1 + v_2) = f(v_1) + f(v_2)$ and $f(kv) = kf(v)$; an *isomorphism* if also bijective ($V \simeq V'$); an *endomorphism* if $V = V'$; an *automorphism* if both. Notation: $\mathrm{Hom}_K(V, V')$, $\mathrm{End}_K(V)$, $\mathrm{Aut}_K(V)$. A linear map is a group homomorphism of $(V, +)$, so $f(0) = 0'$ and $f(-v) = -f(v)$.

Theorem (characterization). C04 · proof: course f is K-linear $\iff$ $f(k_1 v_1 + k_2 v_2) = k_1 f(v_1) + k_2 f(v_2)$, $\forall k_1, k_2 \in K$, $\forall v_1, v_2 \in V$.

Proof. "$\Longrightarrow$" $f(k_1 v_1 + k_2 v_2) = f(k_1 v_1) + f(k_2 v_2) = k_1 f(v_1) + k_2 f(v_2)$. "$\Longleftarrow$" $k_1 = k_2 = 1$ gives additivity; $k_2 = 0$ gives $f(k_1 v_1) = k_1 f(v_1) + 0 \cdot f(v_2) = k_1 f(v_1)$. ∎

What to check — "f is linear". Compute $f(k_1 v_1 + k_2 v_2)$ by the formula of f, regroup, and recognize $k_1 f(v_1) + k_2 f(v_2)$. **Quick test for "no":** if $f(0) \neq 0$ (a constant term, as in $x + a$), f is not linear; products xy, squares, $|x|$, $\sin x$ also break linearity. In $K^n \to K^m$: f is linear exactly when every component is of the form $a_1 x_1 + \cdots + a_n x_n$ (Ex. 7).

Theorem. C04 · proof: homework (book Thm 2.4.5) (i) If $f : V \to V'$ is an isomorphism, then $f^{-1} : V' \to V$ is an isomorphism. (ii) If $f : V \to V'$, $g : V' \to V''$ are linear, then $g \circ f$ is linear.

Proof. (i) f^{-1} is bijective (its inverse is f). Let $k_1, k_2 \in K$, $v_1', v_2' \in V'$ and put $v_i = f^{-1}(v_i')$, so $f(v_i) = v_i'$. By linearity of f, $k_1 v_1' + k_2 v_2' = k_1 f(v_1) + k_2 f(v_2) = f(k_1 v_1 + k_2 v_2)$. Apply f^{-1}: $f^{-1}(k_1 v_1' + k_2 v_2') = k_1 v_1 + k_2 v_2 = k_1 f^{-1}(v_1') + k_2 f^{-1}(v_2')$.

(ii) $(g \circ f)(k_1 v_1 + k_2 v_2) = g\big(k_1 f(v_1) + k_2 f(v_2)\big) = k_1 g(f(v_1)) + k_2 g(f(v_2)) = k_1 (g \circ f)(v_1) + k_2 (g \circ f)(v_2)$. ∎

Kernel and image

Definition. For a linear $f : V \to V'$: $\operatorname{Ker} f = \{v \in V \mid f(v) = 0'\}$ (kernel, null space), $\operatorname{Im} f = \{f(v) \mid v \in V\}$ (image, range).

Theorem. C04 · proof: course $\operatorname{Ker} f \leq V$ and $\operatorname{Im} f \leq V'$.

Proof. $f(0) = 0' \implies 0 \in \operatorname{Ker} f$. For $v_1, v_2 \in \operatorname{Ker} f$: $f(k_1 v_1 + k_2 v_2) = k_1 f(v_1) + k_2 f(v_2) = k_1 \cdot 0' + k_2 \cdot 0' = 0' \implies k_1 v_1 + k_2 v_2 \in \operatorname{Ker} f$.

$0' = f(0) \in \operatorname{Im} f$. For $v_1' = f(v_1)$, $v_2' = f(v_2) \in \operatorname{Im} f$: $k_1 v_1' + k_2 v_2' = k_1 f(v_1) + k_2 f(v_2) = f(k_1 v_1 + k_2 v_2) \in \operatorname{Im} f$. Both by the characterization of subspaces. ∎

Theorem. C04 · proof: course $\operatorname{Ker} f = \{0\} \iff f$ is injective.

Proof. "$\implies$" If $f(v_1) = f(v_2)$, then $f(v_1 - v_2) = f(v_1) - f(v_2) = 0'$, so $v_1 - v_2 \in \operatorname{Ker} f = \{0\}$, i.e. $v_1 = v_2$.

"$\impliedby$" $\{0\} \subseteq \operatorname{Ker} f$ always. If $v \in \operatorname{Ker} f$, then $f(v) = 0' = f(0)$, and injectivity gives $v = 0$. ∎

Theorem. C04 · proof: course $f(\langle X \rangle) = \langle f(X) \rangle$ for every $X \subseteq V$.

Proof. If $X = \emptyset$: $f(\langle \emptyset \rangle) = f(\{0\}) = \{0'\} = \langle \emptyset \rangle = \langle f(\emptyset) \rangle$. If $X \neq \emptyset$, by the characterization of $\langle X \rangle$ and linearity, $f(\langle X \rangle) = \{f(k_1 v_1 + \cdots + k_n v_n)\} = \{k_1 f(v_1) + \cdots + k_n f(v_n) \mid k_i \in K,\ v_i \in X\} = \langle f(X) \rangle$. ∎

Consequence used all the time. If $V = \langle v_1, \ldots, v_n \rangle$, then $\operatorname{Im} f = f(V) = \langle f(v_1), \ldots, f(v_n) \rangle$: the image is generated by the images of a system of generators (e.g. of the canonical basis).

Theorem. C04 · proof: homework (book Thm 2.4.10) $\operatorname{Hom}_K(V, V')$, with $(f + g)(v) = f(v) + g(v)$ and $(kf)(v) = kf(v)$, is a K-vector space. In particular $\operatorname{End}_K(V)$ is a vector space.

Proof. *The operations stay in* $\operatorname{Hom}_K(V, V')$*:* for f, g linear, $(f + g)(k_1 v_1 + k_2 v_2) = f(k_1 v_1 + k_2 v_2) + g(k_1 v_1 + k_2 v_2) = k_1 f(v_1) + k_2 f(v_2) + k_1 g(v_1) + k_2 g(v_2) = k_1 (f + g)(v_1) + k_2 (f + g)(v_2)$, and $(kf)(k_1 v_1 + k_2 v_2) = k\big(k_1 f(v_1) + k_2 f(v_2)\big) = k_1 (kf)(v_1) + k_2 (kf)(v_2)$ (using (L3) and the commutativity of K).

Axioms: $\mathrm{Hom}_K(V,V')$ is a subset of the vector space $(V')^V$ of all functions $V \to V'$ with pointwise operations (a vector space exactly as in Seminar 4, Ex. 3, with K replaced by V'). It contains the zero map and, by the previous paragraph, is closed under linear combinations, so it is a subspace, hence a vector space. ∎

Chapter 5 — Solutions (Seminar 5)

Ex. 1

(i) $\langle 1, X, X^2\rangle = \{a_0 \cdot 1 + a_1 X + a_2 X^2 \mid a_i \in \mathbb{R}\} = \mathbb{R}_2[X]$, the polynomials of degree ≤ 2.

(ii) $a\begin{pmatrix}1&0\\0&0\end{pmatrix} + b\begin{pmatrix}0&1\\0&0\end{pmatrix} + c\begin{pmatrix}0&0\\1&0\end{pmatrix} + d\begin{pmatrix}0&0\\0&1\end{pmatrix} = \begin{pmatrix}a&b\\c&d\end{pmatrix}$, which is an arbitrary matrix.

$\Longrightarrow$ the generated subspace is all of $M_2(\mathbb{R})$.

Ex. 2

(i) $A = \{(0,y,z) \mid y,z \in \mathbb{R}\} = \{y(0,1,0) + z(0,0,1)\} = \langle (0,1,0), (0,0,1)\rangle$.

(ii) $x = -y - z$: $B = \{(-y-z, y, z)\} = \{y(-1,1,0) + z(-1,0,1)\} = \langle (-1,1,0), (-1,0,1)\rangle$.

(iii) $C = \{(x,x,x)\} = \{x(1,1,1)\} = \langle (1,1,1)\rangle$.

Minimality. $C \neq \{0\} = \langle \emptyset \rangle$, so C needs at least one generator. One vector v generates the line $\langle v\rangle = \{tv \mid t \in \mathbb{R}\}$, in which any two vectors are proportional. A contains $(0,1,0)$ and $(0,0,1)$, B contains $(-1,1,0)$ and $(-1,0,1)$ — not proportional — so neither A nor B is generated by one vector.

$\Longrightarrow$ minimal numbers of generators: A: 2, B: 2, C: 1 (= their dimensions, Chapter 7).

Ex. 3

We look for c, d, e as linear combinations of $a = (-2,1,3)$, $b = (3,-2,-1)$:

$$a + b = (1,-1,2) = c, \qquad a - b = (-5,3,4) = d, \qquad 3a - b = (-9,5,10) = e.$$

So $c, d, e \in \langle a, b\rangle \Longrightarrow \langle c,d,e\rangle \subseteq \langle a,b\rangle$ ($\langle a,b\rangle$ is a subspace containing c,d,e).

Conversely, from $c = a + b$, $d = a - b$: $a = \frac{1}{2}c + \frac{1}{2}d$, $b = \frac{1}{2}c - \frac{1}{2}d$, so $a, b \in \langle c,d\rangle \subseteq \langle c,d,e\rangle \Longrightarrow \langle a,b\rangle \subseteq \langle c,d,e\rangle$.

$\Longrightarrow \langle a,b\rangle = \langle c,d,e\rangle$ (a plane through the origin; $e = c + 2d$ is redundant).

How to find the coefficients: solve $\alpha a + \beta b = c$ componentwise, e.g. $-2\alpha + 3\beta = 1$, $\alpha - 2\beta = -1$ $\Longrightarrow \alpha = \beta = 1$, and check the third component $3\alpha - \beta = 2$ ✓.

Ex. 4

$S \leq \mathbb{R}^3$ and $T \leq \mathbb{R}^3$: Seminar 4, Ex. 7 (D and F).

$S \cap T = \{0\}$: if $v = (x,y,z) \in S \cap T$, then $x = y = z$ and $x + y + z = 3x = 0$, so $v = (0,0,0)$.

$S + T = \mathbb{R}^3$: for an arbitrary (x,y,z), let $m = \frac{x+y+z}{3}$. Then

$$(x,y,z) = \underbrace{(x-m,\ y-m,\ z-m)}_{\in S:\ \text{sum}=x+y+z-3m=0} + \underbrace{(m,m,m)}_{\in T}.$$

$\Longrightarrow \mathbb{R}^3 = S \oplus T$**.**

How to find m: we want $(x,y,z)-(m,m,m) \in S$, i.e. $x+y+z-3m=0$.

Ex. 5

$S = \{f \mid f(-x) = f(x)\ \forall x\}$, $T = \{f \mid f(-x) = -f(x)\ \forall x\}$.

Subspaces: the zero function is even and odd. If $f, g \in S$: $(k_1 f + k_2 g)(-x) = k_1 f(-x) + k_2 g(-x) = k_1 f(x) + k_2 g(x) = (k_1 f + k_2 g)(x)$, so $k_1 f + k_2 g \in S$. Same for T with a minus sign.

$S \cap T = \{0\}$**:** if f is even and odd, then $f(x) = f(-x) = -f(x)$, so $2f(x) = 0$, $f(x) = 0$ for all x.

$S + T = \mathbb{R}^{\mathbb{R}}$**:** every f can be written as

$$f(x) = \underbrace{\frac{f(x)+f(-x)}{2}}_{f_e(x)} + \underbrace{\frac{f(x)-f(-x)}{2}}_{f_o(x)},$$

with $f_e(-x) = \frac{f(-x)+f(x)}{2} = f_e(x)$ (even) and $f_o(-x) = \frac{f(-x)-f(x)}{2} = -f_o(x)$ (odd).

$\Longrightarrow \mathbb{R}^{\mathbb{R}} = S \oplus T$**.**

Example: $e^x = \cosh x + \sinh x$.

Ex. 6

Let $k_1, k_2 \in \mathbb{R}$, $v_1 = (x_1, y_1)$, $v_2 = (x_2, y_2)$, so $k_1 v_1 + k_2 v_2 = (k_1 x_1 + k_2 x_2,\ k_1 y_1 + k_2 y_2)$.

$$\begin{aligned} f(k_1 v_1 + k_2 v_2) &= \big((k_1 x_1 + k_2 x_2) + (k_1 y_1 + k_2 y_2),\ (k_1 x_1 + k_2 x_2) - (k_1 y_1 + k_2 y_2)\big) \\ &= k_1(x_1 + y_1,\ x_1 - y_1) + k_2(x_2 + y_2,\ x_2 - y_2) = k_1 f(v_1) + k_2 f(v_2). \end{aligned}$$

The same regrouping works for g and h, because each of their components is of the form $\alpha x + \beta y (+\gamma z)$ (see Ex. 7(i)): e.g. for h, the first component of $h(k_1 v_1 + k_2 v_2)$ is $(k_1 x_1 + k_2 x_2) - (k_1 y_1 + k_2 y_2) = k_1(x_1 - y_1) + k_2(x_2 - y_2)$, and similarly for the others.

$\Longrightarrow f, g \in \mathrm{End}_{\mathbb{R}}(\mathbb{R}^2)$**,** $h \in \mathrm{End}_{\mathbb{R}}(\mathbb{R}^3)$**.**

Ex. 7

(i) Yes, for all a, b, c, d**.** With the notation of Ex. 6:

$$\begin{aligned} f(k_1 v_1 + k_2 v_2) &= \big(a(k_1 x_1 + k_2 x_2) + b(k_1 y_1 + k_2 y_2),\ c(k_1 x_1 + k_2 x_2) + d(k_1 y_1 + k_2 y_2)\big) \\ &= k_1(a x_1 + b y_1,\ c x_1 + d y_1) + k_2(a x_2 + b y_2,\ c x_2 + d y_2) = k_1 f(v_1) + k_2 f(v_2). \end{aligned}$$

(ii) Only for $a = b = 0$**.** If g is linear, then $g(0,0) = (0,0)$; but $g(0,0) = (a,b)$, so $a = b = 0$. Conversely, for $a = b = 0$, $g = 1_{\mathbb{R}^2}$ is linear.

$\Longrightarrow g$ **(a translation) is an endomorphism** $\iff a = b = 0$**.**

Ex. 8

$f(x,y) = (\alpha x + \beta y,\ \gamma x + \delta y)$ with the constants $\alpha = \cos a$, $\beta = -\sin a$, $\gamma = \sin a$, $\delta = \cos a$. By Ex. 7(i), f is linear.

$\Longrightarrow f \in \mathrm{End}_{\mathbb{R}}(\mathbb{R}^2)$ — the rotation of angle a around the origin. It is even an automorphism: its inverse is the rotation of angle $-a$.

Ex. 9

$f(x,y) = (x+y,\ x-y)$. $\mathrm{Ker}\, f$: $x+y=0,\ x-y=0 \Longrightarrow x=y=0 \Longrightarrow \mathrm{Ker}\, f = \{(0,0)\}$. $\mathrm{Im}\, f$: for any (u,v), the system $x+y=u,\ x-y=v$ has the solution $x=\frac{u+v}{2},\ y=\frac{u-v}{2} \Longrightarrow \mathrm{Im}\, f = \mathbb{R}^2$. ($f$ is an automorphism.)

$g(x,y) = (2x-y,\ 4x-2y) = (2x-y)(1,2)$. $\mathrm{Ker}\, g$: $2x-y=0$ (the second equation is twice the first) $\Longrightarrow \mathrm{Ker}\, g = \{(x,2x)\} = \langle(1,2)\rangle$. $\mathrm{Im}\, g = \{t(1,2) \mid t \in \mathbb{R}\} = \langle(1,2)\rangle$, since $t = 2x-y$ takes every real value.

Curiosity: here $\mathrm{Ker}\, g = \mathrm{Im}\, g$, so $g \circ g = 0$.

$h(x,y,z) = (x-y,\ y-z,\ z-x)$. $\mathrm{Ker}\, h$: $x=y,\ y=z \Longrightarrow \mathrm{Ker}\, h = \{(x,x,x)\} = \langle(1,1,1)\rangle$.

$\mathrm{Im}\, h$: every image (u,v,w) satisfies $u+v+w = (x-y)+(y-z)+(z-x) = 0$. Conversely, if $u+v+w=0$, take $(x,y,z) = (u+v,\ v,\ 0)$: $h(u+v,v,0) = (u,\ v,\ -u-v) = (u,v,w)$.

$\Longrightarrow \mathrm{Im}\, h = \{(u,v,w) \mid u+v+w=0\} = \langle(1,0,-1),(0,1,-1)\rangle$.

Check with the First Dimension Theorem (Chapter 7): $2 = 0+2$ for f, $2 = 1+1$ for g, $3 = 1+2$ for h.

Ex. 10

$f(0) = 0 \Longrightarrow 0 \in S$. Let $x_1, x_2 \in S$ ($f(x_i) = x_i$) and $k_1, k_2 \in K$. By linearity:

$$f(k_1x_1 + k_2x_2) = k_1f(x_1) + k_2f(x_2) = k_1x_1 + k_2x_2,$$

so $k_1x_1 + k_2x_2 \in S$.

$\Longrightarrow S \leq V$.

Shorter: $S = \mathrm{Ker}(f - 1_V)$, and $f - 1_V \in \mathrm{End}_K(V)$ because $\mathrm{End}_K(V)$ is a vector space; a kernel is a subspace.

Results at a glance — Seminar 5

Ex.	Answer
1	$\mathbb{R}_2[X]$; $M_2(\mathbb{R})$
2	$A = \langle(0,1,0),(0,0,1)\rangle$, $B = \langle(-1,1,0),(-1,0,1)\rangle$, $C = \langle(1,1,1)\rangle$
3	$c = a+b$, $d = a-b$, $e = 3a-b$; $a = \frac{c+d}{2}$, $b = \frac{c-d}{2}$
4	$m = \frac{x+y+z}{3}$
5	$f = \frac{f(x)+f(-x)}{2} + \frac{f(x)-f(-x)}{2}$
7	(i) always; (ii) iff $a = b = 0$
9	$\mathrm{Ker}\, f = 0$, $\mathrm{Im}\, f = \mathbb{R}^2$; $\mathrm{Ker}\, g = \mathrm{Im}\, g = \langle(1,2)\rangle$; $\mathrm{Ker}\, h = \langle(1,1,1)\rangle$, $\mathrm{Im}\, h = \{u+v+w=0\}$
10	$S = \mathrm{Ker}(f - 1_V)$

Chapter 6 — Linear independence and bases: theory

Course 5 · book §2.6–2.7. From the section on bases on, every vector space is finitely generated.

Linear independence

Definition. A list $(v_1, \dots, v_n)$ (or a set $\{v_1, \dots, v_n\}$) of vectors of V is *linearly independent* if

$$k_1 v_1 + \cdots + k_n v_n = 0 \implies k_1 = \cdots = k_n = 0 \qquad (\forall k_i \in K),$$

and *linearly dependent* otherwise, i.e. if there are $k_1, \dots, k_n$ **not all zero** with $k_1 v_1 + \cdots + k_n v_n = 0$ (a *dependence relation*). An infinite set is independent if every finite subset is.

What to check — "$v_1, \dots, v_n$ are linearly independent". Write $k_1 v_1 + \cdots + k_n v_n = 0$, turn it into a homogeneous linear system in $k_1, \dots, k_n$ (one equation per component / coefficient / entry), and show the only solution is 0. For n vectors in K^n: compute the determinant of their components ($\neq 0 \iff$ independent). For dependence, exhibit one non-zero solution.

Immediate remarks C05 (1) One vector v is dependent $\iff v = 0$ (by $kv = 0 \iff k = 0$ or $v = 0$). (2) A subset of an independent list is independent; a list containing a dependent sublist — in particular, containing 0 (since $1 \cdot 0 = 0$) — is dependent. (3) If $v_1, \dots, v_k$ are independent and $v \notin \langle v_1, \dots, v_k \rangle$, then $v_1, \dots, v_k, v$ are independent (if $c_1 v_1 + \cdots + c_k v_k + cv = 0$ with $c \neq 0$, then $v = -c^{-1}(c_1 v_1 + \cdots + c_k v_k) \in \langle v_1, \dots, v_k \rangle$; so $c = 0$, and then all $c_i = 0$).

Theorem (dependence). C05 · proof: course $v_1, \dots, v_n$ are linearly dependent $\iff$ one of them is a linear combination of the others: $\exists j$, $v_j = \sum_{i \neq j} \alpha_i v_i$.

Proof. "$\implies$" Let $k_1 v_1 + \cdots + k_n v_n = 0$ with some $k_j \neq 0$. Then $-k_j v_j = \sum_{i \neq j} k_i v_i$; multiply by $-k_j^{-1}$ (exists, K is a field): $v_j = \sum_{i \neq j} (-k_j^{-1} k_i) v_i$. Take $\alpha_i = -k_j^{-1} k_i$.

"$\impliedby$" From $v_j = \sum_{i \neq j} \alpha_i v_i$ we get $(-1) v_j + \sum_{i \neq j} \alpha_i v_i = 0$, a linear combination equal to 0 with the coefficient $-1 \neq 0$. ∎

Theorem (dependence in K^n). C05 · proof: course Let $n \geq 2$. (i) Two vectors of K^n are dependent $\iff$ one is a scalar multiple of the other (components proportional). (ii) n vectors of K^n are dependent $\iff$ the determinant of their components is 0.

Proof. (i) By the previous theorem with two vectors: v, v' dependent $\iff v' = kv$ or $v = kv'$, i.e. $x_i' = k x_i$ for all i (or the other way round).

(ii) Write $v_j = (x_{1j}, \dots, x_{nj})$ (components in column j). Then $k_1 v_1 + \cdots + k_n v_n = 0$ is the homogeneous system

$$x_{i1} k_1 + x_{i2} k_2 + \cdots + x_{in} k_n = 0, \qquad i = 1, \dots, n,$$

with unknowns $k_1, \dots, k_n$ and matrix (x_{ij}). The vectors are dependent $\iff$ this system has

a non-zero solution $\iff$ its determinant is 0 (a square homogeneous system has only the zero solution exactly when its determinant is non-zero — Cramer; Chapter 3 of the course).

∎

Geometry. In V_2/V_3: two vectors are dependent iff collinear, three iff coplanar; three vectors in the plane, or four in space, are always dependent.

Bases and coordinates

Definition. A list $B = (v_1, \dots, v_n)$ is a *basis* of V if (i) B is linearly independent and (ii) B is a system of generators: $\langle B \rangle = V$. If $v = k_1 v_1 + \cdots + k_n v_n$, the scalars $k_1, \dots, k_n$ are the *coordinates* of v in B.

Theorem (existence of a basis). C05 · proof: course Every (finitely generated) vector space has a basis.

Proof. If $V = \{0\}$, the empty list is a basis. Otherwise $V = \langle B \rangle$ with $B = (v_1, \dots, v_n)$. If B is independent, it is a basis. If not, by the dependence theorem some v_{j_1} is a linear combination of the others; then every linear combination of B can be rewritten without v_{j_1} (substitute its expression), so $V = \langle B \setminus \{v_{j_1}\} \rangle$. Repeat: as long as the current list is dependent, remove a vector that is a combination of the others — the list still generates V. The process stops after at most $n - 1$ removals: if we reached one vector, $V = \langle v_{j_n} \rangle$, then $v_{j_n} \neq 0$ (else $V = \{0\}$), so this one-element list is independent. In all cases we end with an independent system of generators, i.e. a basis. ∎

Key idea. A dependent system of generators can be **shrunk** without losing generation; stop when it becomes independent. (Dually, Chapter 7: an independent list can be **extended** to a basis.)

Theorem (characterization of a basis). C05 · proof: course $B = (v_1, \dots, v_n)$ is a basis of V $\iff$ every $v \in V$ can be written **uniquely** as $v = k_1 v_1 + \cdots + k_n v_n$.

Proof. "$\Longrightarrow$" Existence: $\langle B \rangle = V$. Uniqueness: if $v = \sum k_i v_i = \sum k'_i v_i$, subtracting gives $\sum (k_i - k'_i) v_i = 0$, and independence gives $k_i = k'_i$ for all i.

"$\Longleftarrow$" Existence of the writings means $\langle B \rangle = V$. If $\sum k_i v_i = 0 = \sum 0 \cdot v_i$, uniqueness of the writing of 0 gives $k_i = 0$ for all i: B is independent. ∎

Recipe — "show B is a basis and find the coordinates of v". Solve $k_1 v_1 + \cdots + k_n v_n = v$ for a **general** v. If the system always has exactly one solution, B is a basis (characterization), and the solution gives the coordinates. For n vectors in K^n the quickest test is $\det \neq 0$ (Course 6).

Standard bases: canonical basis $(e_1, \dots, e_n)$ of K^n (coordinates = components); $(1, X, \dots, X^n)$ of $K_n[X]$ (coordinates = coefficients); (E_{ij}) of $M_{m,n}(K)$ (coordinates = entries); $(1, i)$ of $\mathbb{C}$ over $\mathbb{R}$ (coordinates = real and imaginary part).

Bases and linear maps

Theorem. C05 · proof: course A linear map is determined by its values on a basis: if $B = (v_1, \dots, v_n)$ is a basis of V and $f, g : V \to V'$ are linear with $f(v_i) = g(v_i)$ for all i, then $f = g$.

Proof. For $v \in V$ write $v = \sum k_i v_i$ (unique). Then $f(v) = \sum k_i f(v_i)$ by linearity — so $f(v)$ is determined by the $f(v_i)$; and if $f(v_i) = g(v_i)$, $f(v) = \sum k_i f(v_i) = \sum k_i g(v_i) = g(v)$. ∎

Theorem. C05 · proof: course Let $f : V \to V'$ be linear and $X = (v_1, \dots, v_n)$. (i) f injective, X independent $\Longrightarrow$ $f(X) = (f(v_1), \dots, f(v_n))$ independent. (ii) f surjective, X generates $V \Longrightarrow$ $f(X)$ generates V'. (iii) f bijective, X basis $\Longrightarrow$ $f(X)$ basis of V'.

Proof. (i) If $\sum k_i f(v_i) = 0'$, then $f(\sum k_i v_i) = 0' = f(0)$; injectivity gives $\sum k_i v_i = 0$, and independence of X gives all $k_i = 0$.

(ii) $\langle f(X) \rangle = f(\langle X \rangle) = f(V) = V'$ (the image of a generated subspace, then surjectivity).

(iii) Combine (i) and (ii). ∎

Chapter 6 — Solutions (Seminar 6)

Ex. 1

(i) Three vectors in $\mathbb{R}^3$ — determinant of their components (rows v_1, v_2, v_3):

$$\begin{vmatrix} 1 & -1 & 0 \\ 2 & 1 & 1 \\ 1 & 5 & 2 \end{vmatrix} = 1(2-5) - (-1)(4-1) + 0 = -3 + 3 = 0 \implies \text{dependent.}$$

Dependence relation: $av_1 + bv_2 + cv_3 = 0$ means

$$\begin{cases} a + 2b + c = 0 \\ -a + b + 5c = 0 \\ b + 2c = 0 \end{cases} \implies b = -2c,\ a = -2b - c = 3c.$$

Take $c = 1$:

$\Longrightarrow 3v_1 - 2v_2 + v_3 = 0$ **(check:** $(3 - 4 + 1,\ -3 - 2 + 5,\ 0 - 2 + 2) = (0, 0, 0)$ ✓**).**

(ii) Two vectors are dependent iff proportional. $(2, 1, 1) = k(1, -1, 0)$ would need $k = 2$ and $k = -1$ at the same time — impossible; and $(1, -1, 0) \neq k(2, 1, 1)$ (third component: $0 = k$, then first: $1 = 0$).

$\Longrightarrow v_1, v_2$ **are linearly independent.**

Ex. 2

(i)

$$\begin{vmatrix} 1 & 0 & 2 \\ -1 & 2 & 1 \\ 3 & 1 & 1 \end{vmatrix} = 1(2-1) - 0 + 2(-1-6) = 1 - 14 = -13 \neq 0 \implies \text{independent.}$$

(ii) Add all columns to the first one (the determinant does not change); every row sums to 10 :

$$\begin{vmatrix}1&2&3&4\\2&3&4&1\\3&4&1&2\\4&1&2&3\end{vmatrix} = 10\begin{vmatrix}1&2&3&4\\1&3&4&1\\1&4&1&2\\1&1&2&3\end{vmatrix} = 10\begin{vmatrix}1&2&3&4\\0&1&1&-3\\0&2&-2&-2\\0&-1&-1&-1\end{vmatrix} = 10\begin{vmatrix}1&1&-3\\2&-2&-2\\-1&-1&-1\end{vmatrix}.$$

(second step: subtract row 1 from rows 2–4; third: expand along column 1.) The 3×3 determinant is $1(2-2)-1(-2-2)+(-3)(-2-2)=0+4+12=16$.

$\Longrightarrow$ **determinant** $=160\neq 0 \Longrightarrow v_1,\dots,v_4$ **are linearly independent.**

Ex. 3

$$\begin{vmatrix}1&a&0\\a&1&1\\1&0&a\end{vmatrix} = 1\cdot(a-0)-a(a^2-1)+0 = 2a-a^3 = -a(a^2-2).$$

Independent $\iff$ determinant $\neq 0 \iff a\neq 0$ and $a^2\neq 2$.

$\Longrightarrow$ **$a\in\mathbb{R}\setminus\{0,\sqrt{2},-\sqrt{2}\}$.**

Ex. 4

Three vectors in $\mathbb{R}^4$: no square determinant, so solve $k_1v_1+k_2v_2+k_3v_3=0$ componentwise:

$$\begin{cases}k_1+2k_2=0\\-2k_1+k_2+ak_3=0\\k_2+k_3=0\\-k_1+2k_3=0\end{cases}$$

From equations 1 and 3: $k_1=-2k_2$, $k_3=-k_2$. Then equation 4 holds automatically: $2k_2-2k_2=0$. Equation 2 becomes $4k_2+k_2-ak_2=(5-a)k_2=0$.

If $a\neq 5$: $k_2=0$, hence all $k_i=0$ — independent. If $a=5$: $k_2=1$ gives the non-zero solution $(-2,1,-1)$.

$\Longrightarrow$ **dependent** $\iff a=5$**; then** $-2v_1+v_2-v_3=0$**.**

Equivalently: the 3×4 matrix of components has rank <3 iff all its 3×3 minors vanish; they are $5-a$, $2(5-a)$, 0, $a-5$.

Ex. 5

Solve $av_1+bv_2+cv_3=(x,y,z)$ for a general vector:

$$\begin{cases}a-b+c=x\\a+c=y\\2b+c=z\end{cases} \Longrightarrow b=y-x,\quad c=z-2b=2x-2y+z,\quad a=y-c=-2x+3y-z.$$

(i) For every (x,y,z) the solution exists and is unique $\Longrightarrow (v_1,v_2,v_3)$ is a basis (characterization of a basis). Quick check: $\det\begin{pmatrix}1&1&0\\-1&0&2\\1&1&1\end{pmatrix}=1\neq 0$.

Coordinates of (x,y,z) **in** (v_1,v_2,v_3)**:** $(-2x+3y-z,\ -x+y,\ 2x-2y+z)$**.**

(ii) Plug in e_1, e_2, e_3:

$$e_1 = -2v_1 - v_2 + 2v_3, \qquad e_2 = 3v_1 + v_2 - 2v_3, \qquad e_3 = -v_1 + v_3.$$

(check e_3: $-(1,1,0) + (1,1,1) = (0,0,1)$ ✓)

(iii) In the canonical basis, the coordinates of u are its components: $(1,-1,2)$. In (v_1,v_2,v_3), with $x=1, y=-1, z=2$:

$\Longrightarrow$ $u = -7v_1 - 2v_2 + 6v_3$**, coordinates** $(-7,-2,6)$ **(check:** $(-7+2+6,\ -7+6,\ -4+6) = (1,-1,2)$ ✓**).**

Ex. 6

Idea: replace the v_k by their differences. Let $w_1 = v_1 = (1,\dots,1)$ and, for $k \geq 2$, $w_k = v_k - v_{k-1}$. The last k components of v_k are $1, 2, \dots, k$ and the others are 1, so

$$w_k = (0,\dots,0,\underbrace{1,\dots,1}_{k-1}) = e_{n-k+2} + \cdots + e_n \quad (k \geq 2), \qquad v_k = w_1 + w_2 + \cdots + w_k.$$

Write the unknown coefficients as $\sum_k c_k v_k = \sum_k d_k w_k$ with $d_k = c_k + c_{k+1} + \cdots + c_n$. Component j of $\sum_k d_k w_k$ is $d_1 + \sum_{k \geq n-j+2} d_k$. So $\sum c_k v_k = (x_1,\dots,x_n)$ is equivalent to

$$x_1 = d_1, \qquad x_j - x_{j-1} = d_{n-j+2} \quad (j = 2,\dots,n),$$

which determines all d_k uniquely, and then $c_k = d_k - d_{k+1}$ ($d_{n+1} = 0$) uniquely. So every vector has a unique writing $\Longrightarrow$ $(v_1,\dots,v_n)$ is a basis of $\mathbb{R}^n$.

Coordinates ($n \geq 2$; for $n = 1$, simply $c_1 = x_1$): with $d_1 = x_1$ and $d_k = x_{n-k+2} - x_{n-k+1}$ for $k \geq 2$,

$$c_1 = x_1 + x_{n-1} - x_n, \qquad c_k = x_{n-k+2} - 2x_{n-k+1} + x_{n-k} \ \ (2 \leq k \leq n-1), \qquad c_n = x_2 - x_1.$$

Check $n=3$: $v_1=(1,1,1)$, $v_2=(1,1,2)$, $v_3=(1,2,3)$; $c_1 = x_1 + x_2 - x_3$, $c_2 = x_1 - 2x_2 + x_3$, $c_3 = x_2 - x_1$. For $x = v_3 = (1,2,3)$: $(0,0,1)$ ✓. For $n=2$: $c_1 = 2x_1 - x_2$, $c_2 = x_2 - x_1$, and $(2x_1 - x_2)(1,1) + (x_2 - x_1)(1,2) = (x_1, x_2)$ ✓.

Ex. 7

$(E_1,\dots,E_4)$**:** $\begin{pmatrix} a & b \\ c & d \end{pmatrix} = aE_1 + bE_2 + cE_3 + dE_4$, and the coefficients are forced (they are the entries) $\Longrightarrow$ unique writing $\Longrightarrow$ basis. Coordinates of B: $(2,1,1,0)$.

$(A_1,\dots,A_4)$**:** solve $k_1A_1 + k_2A_2 + k_3A_3 + k_4A_4 = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, entry by entry:

$$\begin{cases} k_1 + k_2 + k_3 + k_4 = a & (1,1) \\ k_2 + k_3 + k_4 = b & (1,2) \\ k_3 + k_4 = c & (2,1) \\ k_1 + k_4 = d & (2,2) \end{cases} \Longrightarrow\ k_1 = a - b,\ \ k_2 = b - c,\ \ k_4 = d - a + b,\ \ k_3 = c - k_4 = a - b + c - d.$$

Unique solution for every matrix $\Longrightarrow$ basis.

For B ($a=2, b=1, c=1, d=0$): $k_1 = 1$, $k_2 = 0$, $k_3 = 2$, $k_4 = -1$.

$\Longrightarrow$ $B = A_1 + 2A_3 - A_4$**, coordinates** $(1,0,2,-1)$ **(check: entry** $(2,2)$**:** $1 - 1 = 0$ ✓**).**

Ex. 8

$E = (1, X, X^2)$**:** a polynomial of degree ≤ 2 is uniquely $a_0 \cdot 1 + a_1 X + a_2 X^2$ (two polynomials are equal iff their coefficients are) $\Longrightarrow$ basis; coordinates (a_0, a_1, a_2).

$B = (1, X - a, (X - a)^2)$**.** *Generators:* write $X = (X - a) + a$:

$$f = a_0 + a_1\big((X - a) + a\big) + a_2\big((X - a) + a\big)^2 = (a_0 + a_1 a + a_2 a^2) + (a_1 + 2a_2 a)(X - a) + a_2 (X - a)^2.$$

Independent: if $c_0 + c_1(X - a) + c_2(X - a)^2 = 0$, the coefficient of X^2 gives $c_2 = 0$, then that of X gives $c_1 = 0$, then $c_0 = 0$.

$\Longrightarrow$ **B is a basis; the coordinates of f are** $\big(a_0 + a_1 a + a_2 a^2,\ \ a_1 + 2a_2 a,\ \ a_2\big) = \big(f(a),\ f'(a),\ \frac{f''(a)}{2}\big)$ **(Taylor).**

Ex. 9

$\mathbb{Z}_2^3$ has $2^3 = 8$ vectors and dimension 3; a basis is a list (v_1, v_2, v_3) of independent vectors (3 independent vectors in $\mathbb{Z}_2^3$ always form a basis — Course 6). Build it vector by vector, using remark (3) on independence:

- $v_1 \neq 0$: $8 - 1 = 7$ choices;
- $v_2 \notin \langle v_1 \rangle = \{0, v_1\}$: $8 - 2 = 6$ choices;
- $v_3 \notin \langle v_1, v_2 \rangle = \{0, v_1, v_2, v_1 + v_2\}$ (4 distinct vectors, since v_1, v_2 are independent): $8 - 4 = 4$ choices.

$\Longrightarrow$ $7 \cdot 6 \cdot 4 = 168$ **bases (as ordered lists, as in the course's definition); as unordered sets:** $168/3! = 28$**.**

In general $\mathbb{Z}_2^n$ has $(2^n - 1)(2^n - 2) \cdots (2^n - 2^{n-1})$ ordered bases; for $n = 2$: $3 \cdot 2 = 6$ (book, Project 2.1).

Ex. 10

A matrix $A \in M_3(\mathbb{Z}_2)$ is invertible $\iff \det A \neq 0 \iff$ its three columns are linearly independent in $\mathbb{Z}_2^3 \iff$ the list of its columns is a basis of $\mathbb{Z}_2^3$. A matrix is the same thing as the ordered list of its columns, so the invertible matrices correspond bijectively to the ordered bases of Ex. 9.

$\Longrightarrow$ $|\mathrm{GL}_3(\mathbb{Z}_2)| = 168$**.**

Results at a glance — Seminar 6

Ex.	Answer
1	$\det = 0$; $3v_1 - 2v_2 + v_3 = 0$
2	$\det = -13$; $\det = 160$
3	$a \notin \{0, \pm\sqrt{2}\}$
4	$a = 5$; $-2v_1 + v_2 - v_3 = 0$
5	coords of (x, y, z): $(-2x + 3y - z,\ -x + y,\ 2x - 2y + z)$; $u \to (-7, -2, 6)$
6	$c_1 = x_1 + x_{n-1} - x_n$, $c_k = x_{n-k+2} - 2x_{n-k+1} + x_{n-k}$, $c_n = x_2 - x_1$

7	B: $(2,1,1,0)$ in E; $(1,0,2,-1)$ in A
8	$(a_0 + a_1 a + a_2 a^2,\ a_1 + 2a_2 a,\ a_2)$
9	168 ordered bases (28 as sets)
10	$\lvert \mathrm{GL}_3(\mathbb{Z}_2) \rvert = 168$

Chapter 7 — Dimension: theory

Course 6 · book §2.8–2.9. All vector spaces are finitely generated. This is the most proof-heavy part of the first partial exam.

Steinitz Exchange Theorem

Lemma. book Lemma 2.8.1 If $z \in \langle y_1, \dots, y_n \rangle$, then $\langle y_1, \dots, y_n, z \rangle = \langle y_1, \dots, y_n \rangle$.

Proof. "$\supseteq$" is clear. "$\subseteq$": $z = \sum a_i y_i$, so in any linear combination $\sum b_i y_i + cz = \sum (b_i + ca_i) y_i$ the vector z can be eliminated. ∎

Theorem (Steinitz, Exchange Theorem). C06 · proof: book Thm 2.8.2 Let $X = (x_1, \dots, x_m)$ be a linearly independent list and $Y = (y_1, \dots, y_n)$ a system of generators of V. Then:
(i) $m \le n$;
(ii) m vectors of Y can be replaced by the vectors of X so that we obtain again a system of generators of V.

Proof. Induction on m.

Step $m = 1$. $X = (x_1)$ is independent, so $x_1 \neq 0$; hence $V \neq \{0\}$ and $n \ge 1 = m$. Since Y generates, $x_1 = \sum_{i=1}^{n} k_i y_i$; not all k_i are 0 (else $x_1 = 0$), say $k_j \neq 0$. Then

$$y_j = k_j^{-1} x_1 - \sum_{i \neq j} k_j^{-1} k_i y_i \in \langle y_1, \dots, y_{j-1}, x_1, y_{j+1}, \dots, y_n \rangle.$$

By the Lemma, $V = \langle y_1, \dots, y_n \rangle = \langle y_1, \dots, y_{j-1}, x_1, y_{j+1}, \dots, y_n \rangle$: we exchanged y_j for x_1.

Step $m - 1 \to m$. $(x_1, \dots, x_{m-1})$ is independent (a sublist of an independent list). By the induction hypothesis, $m - 1 \le n$ and, after renumbering the y's,

$$V = \langle x_1, \dots, x_{m-1}, y_m, \dots, y_n \rangle.$$

(i) If $m - 1 = n$, then $V = \langle x_1, \dots, x_{m-1} \rangle$, so x_m is a combination of $x_1, \dots, x_{m-1}$ — contradicting the independence of X. Hence $m - 1 < n$, i.e. $m \le n$.

(ii) Write $x_m = \sum_{i=1}^{m-1} k_i x_i + \sum_{i=m}^{n} k_i y_i$. Some k_j with $j \ge m$ is non-zero (otherwise $x_m \in \langle x_1, \dots, x_{m-1} \rangle$, contradiction); renumber so that $j = m$. Then

$$y_m = k_m^{-1} x_m - \sum_{i=1}^{m-1} k_m^{-1} k_i x_i - \sum_{i=m+1}^{n} k_m^{-1} k_i y_i \in \langle x_1, \dots, x_m, y_{m+1}, \dots, y_n \rangle,$$

and by the Lemma $V = \langle x_1, \dots, x_{m-1}, y_m, \dots, y_n \rangle = \langle x_1, \dots, x_m, y_{m+1}, \dots, y_n \rangle$. So m vectors of Y were replaced by $x_1, \dots, x_m$. ∎

Key idea. Exchange one vector at a time: write the new x in terms of the current generators, pick a **non-zero coefficient of an old** y (it exists by independence of the x's), solve for that y, and drop it. Note: not necessarily the *first* m vectors of Y are replaced.

Dimension

Theorem. C06 · proof: book Thm 2.8.4 Any two bases of a vector space have the same number of elements.

Proof. Let $B = (v_1, \dots, v_m)$ and $B' = (v'_1, \dots, v'_n)$ be bases. B is independent and B' generates, so $m \leq n$ by Steinitz. B' is independent and B generates, so $n \leq m$. Hence $m = n$. ∎

Definition. $\dim_K V$ = the number of vectors of any basis of V; $\dim\{0\} = 0$. Examples: $\dim K^n = n$, $\dim K_n[X] = n+1$, $\dim M_{m,n}(K) = mn$, $\dim_{\mathbb{C}} \mathbb{C} = 1$, $\dim_{\mathbb{R}} \mathbb{C} = 2$; subspaces of $\mathbb{R}^3$ have dimension 0 (origin), 1 (lines), 2 (planes), 3.

Theorem (characterization of dimension). C06 · proof: course, (i)⟺(iii) homework The following are equivalent: (i) $\dim V = n$; (ii) the maximum number of linearly independent vectors in V is n; (iii) the minimum number of generators of V is n.

Proof. (i)⟹(ii) A basis B has n independent vectors; since B generates, every independent list has $\leq n$ vectors (Steinitz).

(ii)⟹(i) Let B be a basis with m vectors. B is independent, so $m \leq n$ by (ii). Take an independent list with n vectors (exists by (ii)); since B generates, $n \leq m$ (Steinitz). So $m = n$.

(i)⟹(iii) A basis is a system of n generators; since it is independent, every system of generators has $\geq n$ vectors (Steinitz: independent ≤ generators).

(iii)⟹(i) Let B be a basis with m vectors. B generates, so $n \leq m$ by (iii). Take a system of n generators; B is independent, so $m \leq n$ (Steinitz). So $m = n$. ∎

Theorem. C06 · proof: course If $\dim V = n$ and $X = (u_1, \dots, u_n)$ has exactly n vectors, then X is linearly independent $\iff$ X is a system of generators (and then a basis).

Proof. Let $B = (v_1, \dots, v_n)$ be a basis.

"⟹" X independent, B generates: by Steinitz (ii), n vectors of B — i.e. *all* of them — can be replaced by $u_1, \dots, u_n$ and we still get a system of generators. So $\langle X \rangle = V$.

"⟸" Suppose X generates but is dependent. Then some u_j is a combination of the others, and by the Lemma $V = \langle u_1, \dots, u_{j-1}, u_{j+1}, \dots, u_n \rangle$ — a system of $n-1$ generators, contradicting "minimum number of generators $= n$". ∎

Corollary. C06 n vectors of K^n form a basis $\iff$ they are linearly independent $\iff$ the determinant of their components is $\neq 0$.

Key idea (the most used fact in exercises). When the number of vectors equals the dimension, check **only one** of the two basis conditions — usually independence via a determinant.

Completion to a basis and complements

Theorem (completion to a basis). C06 · proof: book Thm 2.8.11 Every linearly independent list in V can be completed to a basis of V.

Proof. Let $(u_1, \dots, u_m)$ be independent and $B = (v_1, \dots, v_n)$ a basis. Since B generates, Steinitz gives $m \le n$ and, after renumbering, $V = \langle u_1, \dots, u_m, v_{m+1}, \dots, v_n \rangle$. This is a system of $n = \dim V$ generators, hence (previous theorem) independent, hence a basis containing $u_1, \dots, u_m$. ∎

Corollary. C06 · proof: homework (book Cor. 2.8.14) Let $S \le V$. (i) Every basis of S can be extended to a basis of V. (ii) $\dim S \le \dim V$. (iii) $\dim S = \dim V \iff S = V$.

Proof. (i) A basis of S is an independent list in V; complete it (Theorem).

(ii) By (i), a basis of S is part of a basis of V, so it has at most $\dim V$ vectors.

(iii) "$\Longleftarrow$" clear. "$\Longrightarrow$" Let $(u_1, \dots, u_n)$ be a basis of S, $n = \dim V$. It is independent in V and has $\dim V$ vectors, so it is a basis of V. Then every $v \in V$ is $\sum k_i u_i \in S$. So $V \subseteq S$, i.e. $S = V$. ∎

Theorem (decomposition, existence of a complement). C06 · proof: book Thm 2.8.15 For every $S \le V$ there is $\bar{S} \le V$ with $V = S \oplus \bar{S}$. In particular $\dim V = \dim S + \dim \bar{S}$.

Proof. Let $(u_1, \dots, u_m)$ be a basis of S; complete it to a basis $B = (u_1, \dots, u_m, v_{m+1}, \dots, v_n)$ of V and put $\bar{S} = \langle v_{m+1}, \dots, v_n \rangle$.

$V = S + \bar{S}$: every $v = \sum_{i \le m} k_i u_i + \sum_{i > m} k_i v_i$, the first sum in S, the second in $\bar{S}$.

$S \cap \bar{S} = \{0\}$: if $v = \sum_{i \le m} k_i u_i = \sum_{i > m} k_i v_i$, then $\sum_{i \le m} k_i u_i - \sum_{i > m} k_i v_i = 0$, and since B is independent, all $k_i = 0$, so $v = 0$.

Finally $(v_{m+1}, \dots, v_n)$ is independent (part of B) and generates $\bar{S}$, so $\dim \bar{S} = n - m$. ∎

Definition. A subspace $\bar{S}$ with $V = S \oplus \bar{S}$ is a *complement* of S. It is **not unique**: in $\mathbb{R}^3$, $\langle e_1, e_2 \rangle$ has the complements $\langle e_3 \rangle$ and $\langle (1,1,1) \rangle$.

Recipe — "complete to a basis / find a complement". Find a basis of S; add vectors (usually canonical ones) until you have $\dim V$ vectors with non-zero determinant; the added vectors generate a complement.

Dimension theorems

Theorem. C06 · proof: book Thm 2.9.1 $V \simeq V' \iff \dim V = \dim V'$.

Proof. "$\Longrightarrow$" Let $f : V \to V'$ be an isomorphism and $B = (v_1, \dots, v_n)$ a basis of V. The vectors $f(v_i)$ are pairwise distinct (f injective) and $f(B)$ is a basis of V' (a bijective linear map sends bases to bases — Chapter 6). So $\dim V' = n$.

"$\Longleftarrow$" Let $B = (v_1, \dots, v_n)$, $B' = (v'_1, \dots, v'_n)$ be bases of V, V'. Define $f(k_1 v_1 + \cdots + k_n v_n) = k_1 v'_1 + \cdots + k_n v'_n$ (well defined: coordinates are unique). *Linear:* if $v = \sum k_i v_i$, $w = \sum l_i v_i$, then $\alpha v + \beta w = \sum(\alpha k_i + \beta l_i) v_i$, so $f(\alpha v + \beta w) = \sum(\alpha k_i + \beta l_i) v'_i = \alpha f(v) + \beta f(w)$. *Bijective:* every $v' = \sum k'_i v'_i \in V'$ equals $f(\sum k'_i v_i)$, and the preimage is unique because the coordinates k'_i of v' in B' are unique. So f is an isomorphism. ∎

Theorem (structure). C06 Every V with $\dim V = n$ is isomorphic to K^n. E.g. $K_n[X] \simeq K^{n+1}$, $M_{m,n}(K) \simeq K^{mn}$ (this is why we may compute in $M_2(\mathbb{R})$ as in $\mathbb{R}^4$).

Definition. For a linear $f : V \to V'$: $\operatorname{null}(f) = \dim \operatorname{Ker} f$ (nullity), $\operatorname{rank}(f) = \dim \operatorname{Im} f$ (rank).

Theorem (First Dimension Theorem, rank-nullity). C06 · proof: book Thm 2.9.5 For every linear $f : V \to V'$: $\dim V = \dim \operatorname{Ker} f + \dim \operatorname{Im} f$.

Proof. Let $(u_1, \dots, u_m)$ be a basis of $\operatorname{Ker} f$; complete it to a basis $B = (u_1, \dots, u_m, v_{m+1}, \dots, v_n)$ of V. We prove that $B' = (f(v_{m+1}), \dots, f(v_n))$ is a basis of $\operatorname{Im} f$; then $\dim V = n = m + (n - m) = \dim \operatorname{Ker} f + \dim \operatorname{Im} f$.

B' independent: if $\sum_{i>m} k_i f(v_i) = 0'$, then $f\big(\sum_{i>m} k_i v_i\big) = 0'$, so $\sum_{i>m} k_i v_i \in \operatorname{Ker} f$, i.e. $\sum_{i>m} k_i v_i = \sum_{i \le m} k_i u_i$ for some $k_1, \dots, k_m$. Then $\sum_{i \le m} k_i u_i - \sum_{i>m} k_i v_i = 0$, and independence of B gives all $k_i = 0$.

B' generates $\operatorname{Im} f$*:* let $v' = f(v)$, $v = \sum_{i \le m} k_i u_i + \sum_{i>m} k_i v_i$. Since $f(u_i) = 0'$, $v' = \sum_{i \le m} k_i f(u_i) + \sum_{i>m} k_i f(v_i) = \sum_{i>m} k_i f(v_i)$. ∎

Key idea. Start from a basis of the kernel, complete it; the images of the **added** vectors form a basis of the image. In exercises: compute $\operatorname{Ker} f$, then $\dim \operatorname{Im} f = \dim V - \dim \operatorname{Ker} f$ is free.

Theorem. C06 · proof: course If $\dim V = \dim V'$ and $f : V \to V'$ is linear, then f injective $\iff$ f surjective $\iff$ f isomorphism.

Proof. Injective $\Longrightarrow \operatorname{Ker} f = \{0\} \Longrightarrow \dim \operatorname{Im} f = \dim V = \dim V'$ (First Dimension Theorem) $\Longrightarrow \operatorname{Im} f = V'$ (a subspace of the same dimension is the whole space) $\Longrightarrow$ surjective.

Surjective $\Longrightarrow \dim \operatorname{Im} f = \dim V' = \dim V \Longrightarrow \dim \operatorname{Ker} f = 0 \Longrightarrow \operatorname{Ker} f = \{0\} \Longrightarrow$ injective. Both together give bijective. ∎

Theorem (Second Dimension Theorem, Grassmann). C06 · proof: book Thm 2.9.7 For $S, T \le V$: $\dim S + \dim T = \dim(S \cap T) + \dim(S + T)$. In particular, if $V = S \oplus T$, then $\dim V = \dim S + \dim T$.

Proof. *Step 1:* $\dim(S \times T) = \dim S + \dim T$. With bases $(s_1, \dots, s_m)$ of S and $(t_1, \dots, t_n)$ of T, every (s, t) is uniquely $(s, 0) + (0, t) = \sum k_i (s_i, 0) + \sum k'_j (0, t_j)$, so $((s_1, 0), \dots, (s_m, 0), (0, t_1), \dots, (0, t_n))$ is a basis of the direct product $S \times T$.

Step 2: the map $f : S \times T \to S + T$, $f(s,t) = s + t$ is linear: $f(k_1(s_1,t_1) + k_2(s_2,t_2)) = k_1 s_1 + k_2 s_2 + k_1 t_1 + k_2 t_2 = k_1 f(s_1,t_1) + k_2 f(s_2,t_2)$. It is surjective by the definition of $S + T$: $\operatorname{Im} f = S + T$.

Step 3: kernel. $f(s,t) = 0 \iff t = -s$, and then $s = -t \in T$, so $s \in S \cap T$: $\operatorname{Ker} f = \{(s,-s) \mid s \in S \cap T\}$. The map $g : S \cap T \to \operatorname{Ker} f$, $g(s) = (s,-s)$, is linear and bijective, so $\dim \operatorname{Ker} f = \dim(S \cap T)$.

Step 4. First Dimension Theorem for f: $\dim S + \dim T = \dim(S \times T) = \dim \operatorname{Ker} f + \dim \operatorname{Im} f = \dim(S \cap T) + \dim(S + T)$. If $V = S \oplus T$, then $S \cap T = \{0\}$ and $S + T = V$. ∎

Key idea. Turn the pair (S,T) into one linear map $(s,t) \mapsto s + t$ and apply rank–nullity: image $= S + T$, kernel $\cong S \cap T$.

Chapter 7 — Solutions (Seminar 7)

Ex. 1

A**:** $A = \{(x,y,0)\} = \langle e_1, e_2 \rangle$; e_1, e_2 are independent (not proportional) $\Longrightarrow$ basis (e_1, e_2), $\dim A = 2$.

B**:** $x = -y - z$: $B = \langle (-1,1,0), (-1,0,1) \rangle$ (Seminar 5, Ex. 2); the two vectors are not proportional $\Longrightarrow$ basis $((-1,1,0), (-1,0,1))$, $\dim B = 2$.

C**:** $C = \langle (1,1,1) \rangle$, $(1,1,1) \neq 0 \Longrightarrow$ basis $((1,1,1))$, $\dim C = 1$.

$\Longrightarrow \dim A = \dim B = 2$ **(planes),** $\dim C = 1$ **(line).**

Ex. 2

(i) $0 \in S$. If $\sum_i x_i = 0$ and $\sum_i y_i = 0$, then $\sum_i (k_1 x_i + k_2 y_i) = k_1 \sum x_i + k_2 \sum y_i = 0 \Longrightarrow S \leq K^n$ (it is also $\operatorname{Ker} f$ for the linear map $f(x) = x_1 + \cdots + x_n$).

(ii) From $x_n = -x_1 - \cdots - x_{n-1}$:

$$(x_1, \ldots, x_{n-1}, -x_1 - \cdots - x_{n-1}) = \sum_{i=1}^{n-1} x_i \, (e_i - e_n).$$

So $u_i = e_i - e_n$ $(i = 1, \ldots, n-1)$ generate S. They are independent: $\sum_{i=1}^{n-1} k_i u_i = (k_1, \ldots, k_{n-1}, -\sum k_i) = 0$ forces $k_1 = \cdots = k_{n-1} = 0$.

$\Longrightarrow$ **basis** $(e_1 - e_n, \ldots, e_{n-1} - e_n)$**,** $\dim S = n - 1$**.**

Check with rank–nullity: $f(x) = x_1 + \cdots + x_n$ is surjective onto K, so $\dim S = \dim \operatorname{Ker} f = n - 1$.

Ex. 3

$\mathbb{C}$ **over** $\mathbb{C}$**:** every $z = z \cdot 1$, uniquely $\Longrightarrow$ basis (1), $\dim_{\mathbb{C}} \mathbb{C} = 1$.

$\mathbb{C}$ **over** $\mathbb{R}$**:** every $z = x \cdot 1 + y \cdot i$ with unique $x, y \in \mathbb{R} \Longrightarrow$ basis $(1, i)$, $\dim_{\mathbb{R}} \mathbb{C} = 2$.

$\{1, i\}$ **over** $\mathbb{C}$**:** the scalars may be complex: $i \cdot 1 + (-1) \cdot i = 0$ with non-zero coefficients $\Longrightarrow$ dependent. (Also: two vectors in a 1-dimensional space are always dependent.)

$\{1, i\}$ **over** $\mathbb{R}$**:** $a \cdot 1 + b \cdot i = 0$ with $a, b \in \mathbb{R}$ means $a + bi = 0$, so $a = b = 0 \Longrightarrow$ independent.

Ex. 4

Linear: both components y and $-x$ are of the form $\alpha x + \beta y + \gamma z$, so $f(k_1 v_1 + k_2 v_2) = (k_1 y_1 + k_2 y_2,\ -k_1 x_1 - k_2 x_2) = k_1 f(v_1) + k_2 f(v_2)$.

$\operatorname{Ker} f$**:** $y = 0$, $-x = 0 \Longrightarrow \operatorname{Ker} f = \{(0,0,z)\} = \langle e_3 \rangle$, basis (e_3), $\dim \operatorname{Ker} f = 1$.

$\operatorname{Im} f$**:** for any $(u,v) \in \mathbb{R}^2$, $f(-v,u,0) = (u,v) \Longrightarrow \operatorname{Im} f = \mathbb{R}^2$, basis (e_1, e_2), $\dim \operatorname{Im} f = 2$.

Check: $\dim \mathbb{R}^3 = 3 = 1 + 2$ ✓**.**

Ex. 5

$\operatorname{Ker} f$**:** $-y + 5z = 0$, $x = 0$, $y - 5z = 0 \Longrightarrow x = 0$, $y = 5z \Longrightarrow \operatorname{Ker} f = \{(0, 5z, z)\} = \langle (0,5,1) \rangle$, $\dim \operatorname{Ker} f = 1$.

$\operatorname{Im} f = \langle f(e_1), f(e_2), f(e_3) \rangle = \langle (0,1,0),\ (-1,0,1),\ (5,0,-5) \rangle$. Since $f(e_3) = -5 f(e_2)$, $\operatorname{Im} f = \langle (0,1,0), (-1,0,1) \rangle$, and these two are not proportional $\Longrightarrow$ basis $((0,1,0),(-1,0,1))$, $\dim \operatorname{Im} f = 2$.

Check: $3 = 1 + 2$ ✓**. Equivalently,** $\operatorname{Im} f = \{(u,v,w) \mid u + w = 0\}$**.**

Ex. 6

In $\mathbb{R}^3$ ($\dim = 3$) it is enough to add vectors so that the determinant of the three vectors is non-zero.

A**:** (e_1, e_2) + $e_3 \Longrightarrow$ the canonical basis (e_1, e_2, e_3).

B**:** $((-1,1,0),(-1,0,1))$ + e_1**:** $\begin{vmatrix} -1 & 1 & 0 \\ -1 & 0 & 1 \\ 1 & 0 & 0 \end{vmatrix} = 1 \cdot (1 \cdot 1 - 0 \cdot 0) = 1 \neq 0$ (expansion along row 3) $\Longrightarrow$ basis $((-1,1,0),(-1,0,1),(1,0,0))$.

C**:** $((1,1,1))$ + e_1, e_2**:** $\begin{vmatrix} 1 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{vmatrix} = 1 \neq 0 \Longrightarrow$ basis $((1,1,1), e_1, e_2)$.

The completions are not unique (e.g. A can also be completed with $(1,1,1)$).

Ex. 7

(i) $x = -2y - 3z$: $A = \langle (-2,1,0),(-3,0,1) \rangle$, $\dim A = 2$. Complete with e_1:

$$\begin{vmatrix} -2 & 1 & 0 \\ -3 & 0 & 1 \\ 1 & 0 & 0 \end{vmatrix} = 1 \cdot (1 \cdot 1 - 0 \cdot 0) = 1 \neq 0.$$

$\Longrightarrow \bar{A} = \langle (1,0,0) \rangle$ **is a complement:** $\mathbb{R}^3 = A \oplus \langle e_1 \rangle$ **(any vector not in** A **generates a complement, e.g.** $\langle (1,2,3) \rangle$**).**

(ii) $B = \langle X, X^3 \rangle$ and X, X^3 are independent, $\dim B = 2$, in $\mathbb{R}_3[X]$ with $\dim = 4$. Complete (X, X^3) with $1, X^2$ to the basis $(1, X, X^2, X^3)$.

$\Longrightarrow \bar{B} = \langle 1, X^2 \rangle = \{a + cX^2 \mid a, c \in \mathbb{R}\}$ **(odd-degree part ⊕ even-degree part).**

Ex. 8

Second Dimension Theorem for (S,U) and for (T,U):

$$\dim S = \dim(S \cap U) + \dim(S + U) - \dim U, \qquad \dim T = \dim(T \cap U) + \dim(T + U) - \dim U.$$

The right-hand sides are equal by hypothesis $\Longrightarrow \dim S = \dim T$. Since $S \subseteq T$, S is a subspace of the vector space T with the same dimension, so $S = T$ (Corollary (iii), applied in T).

$\Longrightarrow S = T$.

Ex. 9

$T = \{a(0,1,1) + b(1,1,0)\} = \{(b,\ a+b,\ a) \mid a, b \in \mathbb{R}\}$. A vector of T lies in S iff its first component is 0, i.e. $b = 0$:

$\Longrightarrow S \cap T = \{(0,a,a)\} = \langle (0,1,1) \rangle$, $\dim(S \cap T) = 1$.

$\dim S = 2$ ($S = \langle e_2, e_3 \rangle$), $\dim T = 2$ (the two generators are not proportional). By the Second Dimension Theorem:

$$\dim(S + T) = \dim S + \dim T - \dim(S \cap T) = 2 + 2 - 1 = 3 = \dim \mathbb{R}^3.$$

$\Longrightarrow S + T = \mathbb{R}^3$ **(a subspace of full dimension). The sum is not direct.**

Ex. 10

Use the isomorphism $M_2(\mathbb{R}) \simeq \mathbb{R}^4$, $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto (a,b,c,d)$ (it preserves dimensions and independence):

$S = \langle (1,1,0,0), (1,0,1,1) \rangle$, $T = \langle (0,1,1,0), (0,0,1,1) \rangle$. In each pair the vectors are not proportional $\Longrightarrow \dim S = \dim T = 2$.

$S + T = \langle S \cup T \rangle$ is generated by the four vectors; their determinant:

$$\begin{vmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{vmatrix} \overset{R_2 - R_1}{=} \begin{vmatrix} 1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{vmatrix} = \begin{vmatrix} -1 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix} = -1 - 1 + 1 = -1 \neq 0.$$

So the four vectors are independent $\Longrightarrow \dim(S + T) = 4$, i.e. $S + T = M_2(\mathbb{R})$. Then

$$\dim(S \cap T) = \dim S + \dim T - \dim(S + T) = 2 + 2 - 4 = 0.$$

$\Longrightarrow \dim S = \dim T = 2$, $\dim(S + T) = 4$, $\dim(S \cap T) = 0$; **so** $M_2(\mathbb{R}) = S \oplus T$.

Results at a glance — Seminar 7

Ex.	Answer
1	$\dim A = 2$, $\dim B = 2$, $\dim C = 1$
2	basis $(e_i - e_n)_{i<n}$, $\dim S = n - 1$
3	$\dim_{\mathbb{C}} \mathbb{C} = 1$, $\dim_{\mathbb{R}} \mathbb{C} = 2$
4	$\operatorname{Ker} f = \langle e_3 \rangle$, $\operatorname{Im} f = \mathbb{R}^2$
5	$\operatorname{Ker} f = \langle (0,5,1) \rangle$, $\operatorname{Im} f = \langle (0,1,0), (-1,0,1) \rangle$
6	add e_3; e_1; e_1, e_2
7	$\langle (1,0,0) \rangle$; $\langle 1, X^2 \rangle$

8	Grassmann $\Longrightarrow \dim S = \dim T \Longrightarrow S = T$
9	$S \cap T = \langle (0,1,1) \rangle$, $\dim(S+T) = 3$
10	$2,\ 2,\ 4,\ 0$

Chapter 8 — True or false? Answers with justification

Statements of the workbook, Chapter 8 (based on Crivei's chapter quizzes). For every F there is a concrete counterexample; for every T, the reason.

Quiz 1 — Preliminaries

#		Justification
1	**F**	$R = \{(1,2),(2,1)\}$ on $\{1,2\}$ is symmetric, but $(1,1) \notin R$.
2	**F**	The partition $\{A\}$ has one block; the intersection of its blocks is $A \neq \emptyset$. (True only for partitions with at least two blocks.)
3	**F**	A partition is a set of *subsets* of A, i.e. a subset of $\mathcal{P}(A)$, not of A: $\{\{1\},\{2\}\} \not\subseteq \{1,2\}$.
4	**T**	$F : E(A) \to P(A)$, $r \mapsto A/r$ is a bijection (Chapter 2).
5	**T**	A group is a monoid, and a monoid is a semigroup (associativity is part of the definition).
6	**F**	$1 \cdot x = x \cdot 1$ in every group; also $x \cdot x = x \cdot x$.
7	**F**	$(xy)^{-1} = y^{-1}x^{-1}$; in S_3 with $x = (1\,2)$, $y = (2\,3)$ one gets $y^{-1}x^{-1} \neq x^{-1}y^{-1}$. (True in abelian groups.)
8	**T**	If $1'$ is the identity of H: $1'1' = 1' = 1' \cdot 1$ in G, cancel $\Longrightarrow 1' = 1$.
9	**F**	The inverse in H is also an inverse in G, and inverses in G are unique.
10	**T**	$xy = yx$ holds for all elements of G, in particular for those of H.
11	**F**	$\mathbb{N}$ is stable in $(\mathbb{Z},+)$ but not a subgroup ($-1 \notin \mathbb{N}$).
12	**T**	The trivial homomorphism $f(x) = 1'$: $f(xy) = 1' = 1'1' = f(x)f(y)$.
13	**F**	The trivial homomorphism $\mathbb{Z} \to \mathbb{Z}$, $x \mapsto 0$ is not bijective.
14	**F**	$(R,\cdot)$ is only a semigroup: in $\mathbb{Z}$, 2 has no multiplicative inverse (0 never has one).
15	**F**	$(R,+)$ is abelian in every ring, by definition; e.g. $M_2(\mathbb{R})$ is non-commutative.
16	**T**	$(x+y)(x-y) = x^2 - xy + yx - y^2 = x^2 - y^2$, using $yx = xy$ and the ring rules.
17	**F**	$(K,\cdot)$ is not a group: 0 has no inverse ($0 \cdot x = 0 \neq 1$). Only $(K^*,\cdot)$ is a group.
18	**T**	$(K^*,\cdot)$ is a group, so 1 exists; with $0 \cdot x = 0$, "$\cdot$" is associative on all of K.
19	**T**	A field is by definition a commutative division ring.
20	**F**	$2\mathbb{Z}$ is a subring of $\mathbb{Z}$ without identity ($2x = x$ only for $x = 0$).

21	**T**	A subfield is a field, so it has a multiplicative identity (in fact $1 \in A$, by the characterization).
22	**T**	$f(x+y) = f(x) + f(y)$ is the first condition of a ring homomorphism.
23	**F**	The trivial homomorphism $\mathbb{Z} \to \mathbb{Z}$, $x \mapsto 0$ maps 1 to 0; also $\mathbb{R} \to M_2(\mathbb{R})$, $x \mapsto \begin{pmatrix} x & 0 \\ 0 & 0 \end{pmatrix}$.
24	**F**	$A = B = I_2$: $\det(2I_2) = 4 \neq 2 = \det I_2 + \det I_2$.
25	**F**	$\operatorname{rank} A \leq \min(m, n)$; e.g. the zero 2×3 matrix has rank 0.

Quiz 2 — Vector spaces

#		Justification
1	**F**	It is a function $K \times V \to V$ (scalar times vector).
2	**T**	(L2) gives $k_1 v + k_2 v$, and $(V, +)$ is commutative.
3	**T**	$(k_1 k_2)v = (k_2 k_1)v \overset{(L3)}{=} k_2(k_1 v)$, since K is commutative.
4	**F**	$\mathbb{Z}_2^n$ over $\mathbb{Z}_2$ has 2^n vectors.
5	**T**	For $v \neq 0$, $kv = k'v \Rightarrow (k - k')v = 0 \Rightarrow k = k'$; so the vectors kv, $k \in K$, are pairwise distinct.
6	**F**	$\{(0,0), (1,0)\} \subseteq \mathbb{R}^2$ contains 0 but not $2(1,0)$.
7	**T**	Theorem of Chapter 5: 0 is in all of them, and each is closed under linear combinations.
8	**F**	$\{(x,0)\} \cup \{(0,y)\}$ in $\mathbb{R}^2$: $(1,0) + (0,1) = (1,1)$ is outside.
9	**F**	$\langle \emptyset \rangle = \{0\}$ (the smallest subspace, and every subspace contains 0).
10	**T**	$\sum_j a_j \left(\sum_i k_{ij} v_i \right) = \sum_i \left(\sum_j a_j k_{ij} \right) v_i$.
11	**T**	$s = s + 0 \in S + T$ and $t = 0 + t \in S + T$.
12	**T**	$f(v_1 + v_2) = f(v_1) + f(v_2)$: a homomorphism of $(V, +)$ into $(V', +)$.
13	**T**	Chapter 5 (proof: $k_1 v_1' + k_2 v_2' = f(k_1 v_1 + k_2 v_2)$, apply f^{-1}).
14	**F**	(e_1, e_1, e_2) in $\mathbb{R}^2$ is dependent, but e_2 is not a combination of e_1, e_1. (Only *some* vector is.)
15	**F**	$(e_1, 0)$: $k_1 e_1 + k_2 \cdot 0 = 0$ forces $k_1 = 0$. Only "not all zero" is guaranteed.
16	**F**	$v_1 = v_3 = e_1$, $v_2 = e_2$: (v_1, v_2) and (v_2, v_3) are independent, $(v_1, v_3) = (e_1, e_1)$ is not.
17	**T**	$V = \langle V \rangle$, so V itself is a system of generators (and in the course all spaces are finitely generated).
18	**T**	Lemma of Chapter 7: removing a vector that is a combination of the others does not change the generated subspace.
19	**F**	$((1,0), (0,1))$ and $((1,1), (0,1))$ are two bases of $\mathbb{R}^2$.
20	**F**	$\emptyset$ is a basis of $\{0\}$, so $\dim\{0\} = 0$.
21	**F**	The minimum number of generators is exactly n (characterization of dimension).

22	**F**	The maximum number of independent vectors is n (Steinitz).
23	**F**	$\langle e_1, e_2\rangle \leq \mathbb{R}^3$ has the complements $\langle e_3\rangle$ and $\langle (1,1,1)\rangle$.
24	**F**	(e_1, e_2, e_1+e_2) generates $\mathbb{R}^2$ but is dependent, and adding vectors never makes a list independent. A system of generators can be *reduced* to a basis, not completed.
25	**F**	$\dim S + \dim T = \dim(S\cap T) + \dim(S+T) \geq \dim(S+T)$; e.g. $S = T$ a line: $2 > 1$.

Chapter 9 — Proof practice: solutions

Solutions of the new exercises P1–P13 of the workbook. For the course theorems (9.1 in the workbook), see the theory parts of Chapters 1–7.

P1

(i) $1 \in H$ and $1 \in K \Longrightarrow 1 \in H \cap K$. If $x, y \in H \cap K$, then $xy^{-1} \in H$ (as $H \leq G$) and $xy^{-1} \in K$ (as $K \leq G$) $\Longrightarrow xy^{-1} \in H \cap K$.

$\Longrightarrow H \cap K \leq G$ **(characterization of subgroups).**

(ii) "$\Longleftarrow$" If $H \subseteq K$, then $H \cup K = K \leq G$; similarly if $K \subseteq H$.

"$\Longrightarrow$" Suppose $H \cup K \leq G$ but $H \not\subseteq K$ and $K \not\subseteq H$. Take $h \in H \setminus K$ and $k \in K \setminus H$. Then $hk \in H \cup K$ (closure).

- If $hk \in H$, then $k = h^{-1}(hk) \in H$ — contradiction.
- If $hk \in K$, then $h = (hk)k^{-1} \in K$ — contradiction.

$\Longrightarrow H \subseteq K$ **or** $K \subseteq H$**.**

P2

$\operatorname{Ker} f \leq G$**:** $f(1) = 1' \Longrightarrow 1 \in \operatorname{Ker} f$. For $x, y \in \operatorname{Ker} f$: $f(xy^{-1}) = f(x)f(y)^{-1} = 1' \cdot 1'^{-1} = 1'$ (using $f(y^{-1}) = f(y)^{-1}$) $\Longrightarrow xy^{-1} \in \operatorname{Ker} f$.

$\operatorname{Im} f \leq G'$**:** $1' = f(1) \in \operatorname{Im} f$. For $f(x), f(y) \in \operatorname{Im} f$: $f(x)f(y)^{-1} = f(x)f(y^{-1}) = f(xy^{-1}) \in \operatorname{Im} f$.

Injectivity: "$\Longrightarrow$" if $f(x) = 1' = f(1)$, then $x = 1$. "$\Longleftarrow$" if $f(x) = f(y)$, then $f(xy^{-1}) = f(x)f(y)^{-1} = 1'$, so $xy^{-1} \in \operatorname{Ker} f = \{1\}$, i.e. $x = y$.

This is the group version of the vector space theorems of Chapter 5 — the same proofs, multiplicatively.

P3

$1x = x = x1$ for all $x \Longrightarrow 1 \in Z(G)$.

Inverses: if $b \in Z(G)$, then for all x: $bx = xb$; multiply on both sides by b^{-1} (left and right): $b^{-1}(bx)b^{-1} = b^{-1}(xb)b^{-1} \Longrightarrow xb^{-1} = b^{-1}x$. So $b^{-1} \in Z(G)$.

Products: if $a, c \in Z(G)$, then $(ac)x = a(cx) = a(xc) = (ax)c = (xa)c = x(ac) \Longrightarrow ac \in Z(G)$.

$\Longrightarrow Z(G) \leq G$ **(form (b) of the characterization).** $Z(G) = G \iff G$ **is abelian.**

P4

Operation: if $x, y \in U(R)$, then xy is invertible with inverse $y^{-1}x^{-1}$: $(xy)(y^{-1}x^{-1}) = x(yy^{-1})x^{-1} = 1$ and $(y^{-1}x^{-1})(xy) = 1$. So $xy \in U(R)$.

Associativity: inherited from $(R, \cdot)$. *Identity:* $1 \cdot 1 = 1 \Longrightarrow 1 \in U(R)$. *Inverses:* for $x \in U(R)$, x^{-1} is invertible (its inverse is x), so $x^{-1} \in U(R)$.

$\Longrightarrow$ $(U(R), \cdot)$ **is a group (the group of units).**

$U(\mathbb{Z}) = \{-1, 1\}$ ($xy = 1$ in $\mathbb{Z}$ forces $|x| = 1$). $U(\mathbb{Z}_n) = \{\widehat{a} \mid \gcd(a, n) = 1\}$ (Seminar 3, Ex. 9). For a field, $U(K) = K^*$.

P5

Let $xy = 0$ and suppose $x \neq 0$. Then x^{-1} exists and

$$y = 1 \cdot y = (x^{-1}x)y = x^{-1}(xy) = x^{-1} \cdot 0 = 0.$$

$\Longrightarrow$ **a field has no zero divisors.**

In $\mathbb{Z}_6$: $\widehat{2} \cdot \widehat{3} = \widehat{6} = \widehat{0}$ with $\widehat{2}, \widehat{3} \neq \widehat{0} \Longrightarrow \mathbb{Z}_6$ is not a field.

P6

(r) $ab = ba \Longrightarrow (a, b) \sim (a, b)$. (s) $ad = bc \Longrightarrow cb = da \Longrightarrow (c, d) \sim (a, b)$.

(t) Let $ad = bc$ and $cf = de$. Multiply the first by f: $adf = bcf = b(cf) = b(de)$. So $d(af) = d(be)$, and since $d \neq 0$ (second components are in $\mathbb{Z}^*$), cancellation in $\mathbb{Z}$ gives $af = be$, i.e. $(a, b) \sim (e, f)$.

$\Longrightarrow$ $\sim$ **is an equivalence relation.**

Class of (a, b): all pairs (c, d) with $\frac{c}{d} = \frac{a}{b}$ — the representations of the fraction $\frac{a}{b}$. The quotient set is in bijection with $\mathbb{Q}$ (this is how $\mathbb{Q}$ is constructed from $\mathbb{Z}$).

Where $d \neq 0$ is used: without it, $(1, 0) \sim (0, 0) \sim (0, 1)$ but $(1, 0) \not\sim (0, 1)$.

P7

"$\Longleftarrow$" If $S \subseteq T$, then $S \cup T = T \leq V$; similarly for $T \subseteq S$.

"$\Longrightarrow$" Suppose $S \cup T \leq V$, $S \not\subseteq T$, $T \not\subseteq S$. Take $s \in S \setminus T$, $t \in T \setminus S$. Then $s + t \in S \cup T$.

- If $s + t \in S$, then $t = (s + t) - s \in S$ — contradiction.
- If $s + t \in T$, then $s = (s + t) - t \in T$ — contradiction.

$\Longrightarrow$ $S \subseteq T$ **or** $T \subseteq S$**. (Same idea as P1(ii).)**

P8

$f(0) = 0' \in S' \Longrightarrow 0 \in f^{-1}(S')$. Let $v_1, v_2 \in f^{-1}(S')$, i.e. $f(v_1), f(v_2) \in S'$, and $k_1, k_2 \in K$. Then

$$f(k_1v_1 + k_2v_2) = k_1f(v_1) + k_2f(v_2) \in S'$$

because S' is a subspace. So $k_1v_1 + k_2v_2 \in f^{-1}(S')$.

$\Longrightarrow$ $f^{-1}(S') \leq V$**. (For** $S' = \{0'\}$ **we recover** $\operatorname{Ker} f \leq V$**.)**

P9

$\operatorname{Ker} f \cap \operatorname{Im} f = \{0\}$**:** let $v \in \operatorname{Ker} f \cap \operatorname{Im} f$, $v = f(w)$. Then $0 = f(v) = f(f(w)) = (f \circ f)(w) = f(w) = v$.

$\operatorname{Ker} f + \operatorname{Im} f = V$**:** for every $v \in V$,

$$v = \underbrace{\big(v - f(v)\big)}_{\in \operatorname{Ker} f} + \underbrace{f(v)}_{\in \operatorname{Im} f}, \qquad f\big(v - f(v)\big) = f(v) - f(f(v)) = f(v) - f(v) = 0.$$

$\Longrightarrow V = \operatorname{Ker} f \oplus \operatorname{Im} f$**.**

Example: $f(x, y) = (x, 0)$ on $\mathbb{R}^2$: $\operatorname{Ker} f = \langle e_2 \rangle$, $\operatorname{Im} f = \langle e_1 \rangle$.

P10

Let $u_1 = v_1$, $u_2 = v_1 + v_2$, $u_3 = v_1 + v_2 + v_3$. Since $\dim V = 3$, it is enough to prove independence.

$$au_1 + bu_2 + cu_3 = (a + b + c)v_1 + (b + c)v_2 + cv_3 = 0 \implies c = 0,\ b + c = 0,\ a + b + c = 0$$

(independence of v_1, v_2, v_3) $\Longrightarrow c = b = a = 0$. So (u_1, u_2, u_3) is independent, hence a basis (3 independent vectors in a 3-dimensional space).

Coordinates: $v_1 = u_1$, $v_2 = u_2 - u_1$, $v_3 = u_3 - u_2$.

$\Longrightarrow v_1 \to (1, 0, 0)$**,** $v_2 \to (-1, 1, 0)$**,** $v_3 \to (0, -1, 1)$**.**

P11

First Dimension Theorem: $\dim V = \dim \operatorname{Ker} f + \dim \operatorname{Im} f$, and $\operatorname{Im} f \leq V'$ gives $\dim \operatorname{Im} f \leq \dim V'$.

If $\dim V > \dim V'$: $\dim \operatorname{Ker} f = \dim V - \dim \operatorname{Im} f \geq \dim V - \dim V' > 0 \Longrightarrow \operatorname{Ker} f \neq \{0\} \Longrightarrow f$ is not injective.

If $\dim V < \dim V'$: $\dim \operatorname{Im} f = \dim V - \dim \operatorname{Ker} f \leq \dim V < \dim V' \Longrightarrow \operatorname{Im} f \neq V' \Longrightarrow f$ is not surjective.

$\Longrightarrow$ an isomorphism can exist only between spaces of equal dimension (compare with Chapter 7).

P12

$S + T \leq V \Longrightarrow \dim(S + T) \leq n$. By the Second Dimension Theorem:

$$\dim(S \cap T) = \dim S + \dim T - \dim(S + T) \ \geq\ \dim S + \dim T - n \ >\ 0.$$

$\Longrightarrow S \cap T \neq \{0\}$**.**

Two planes through the origin in $\mathbb{R}^3$: $2 + 2 = 4 > 3 \Longrightarrow \dim(S \cap T) \geq 1$: they share at least a line (exactly a line if they are different, the whole plane if they coincide).

P13

$\operatorname{Ker} f \subseteq \operatorname{Ker}(g \circ f)$**:** $f(v) = 0' \Longrightarrow g(f(v)) = g(0') = 0''$.

$\operatorname{Im}(g \circ f) \subseteq \operatorname{Im} g$**:** $(g \circ f)(v) = g(f(v))$ is an image of g.

Ranks: from the second inclusion, $\operatorname{rank}(g \circ f) \leq \operatorname{rank} g$. Also $\operatorname{Im}(g \circ f) = g(\operatorname{Im} f)$, the image of the linear map $g|_{\operatorname{Im} f} : \operatorname{Im} f \to V''$; by the First Dimension Theorem applied to it, $\dim g(\operatorname{Im} f) \leq \dim \operatorname{Im} f = \operatorname{rank} f$.

$\Longrightarrow \operatorname{rank}(g \circ f) \leq \min(\operatorname{rank} f, \operatorname{rank} g)$.

Alternatively, from the first inclusion: $\operatorname{rank}(g \circ f) = \dim V - \dim \operatorname{Ker}(g \circ f) \leq \dim V - \dim \operatorname{Ker} f = \operatorname{rank} f$.

Exam proofs — quick sheet

Every theorem of Chapters 1–2 of the course (= Chapters 1–7 of this document), reduced to the steps you must write. Revise by covering the right column and reconstructing the proof from the statement. Full proofs: in the theory parts above.

Preliminaries (Courses 1–2)

Statement	Key steps
Identity element is unique	$e_1 = e_1 e_2 = e_2$.
Inverse is unique (associative op.)	$a_1 = a_1(aa_2) = (a_1 a)a_2 = a_2$.
Cancellation; $(x^{-1})^{-1} = x$; $(xy)^{-1} = y^{-1}x^{-1}$	multiply by a^{-1}; x is an inverse of x^{-1}; $(xy)(y^{-1}x^{-1}) = 1 = (y^{-1}x^{-1})(xy)$ + uniqueness.
f has an inverse $\Longleftrightarrow$ f bijective	$\Longrightarrow$ $g \circ f = 1$ gives injective, $f \circ g = 1$ surjective. $\Longleftarrow$ $g(y)$ = the unique preimage. Uniqueness: $g_1 = g_1(fg_2) = (g_1 f)g_2 = g_2$.
$\lvert B^A \rvert = \lvert B \rvert^{\lvert A \rvert}$	induction on $\lvert A \rvert$: restriction to A' (m^n) × value at the new point (m).
Equivalences $\longleftrightarrow$ partitions	(i) (r) $\Longrightarrow$ classes non-empty, cover; $a \in r\langle x \rangle \cap r\langle y \rangle \Longrightarrow x\,r\,y$ (s,t) $\Longrightarrow$ classes equal. (ii) r_π: (r) covering, (s) obvious, (t) common element $\Longrightarrow$ same block. (iii) $G(F(r)) = r$, $F(G(\pi)) = \pi$.
Ring rules: $x \cdot 0 = 0$, $x(-y) = -xy$	$x(y-z) + xz = xy$; $x \cdot 0 = x(y-y) = 0$; $x(-y) + xy = x \cdot 0 = 0$.
Subgroup $\Longleftrightarrow$ $1 \in H$, $xy^{-1} \in H$	$\Longrightarrow$ identity of H: $1'1' = 1'1$, cancel $\Longrightarrow$ $1' = 1$; inverse in H = inverse in G (uniqueness). $\Longleftarrow$ $1 = xx^{-1}$, $y^{-1} = 1y^{-1}$, $xy = x(y^{-1})^{-1}$; associativity inherited.
Subring $\Longleftrightarrow$ $0 \in A$, $x - y$, $xy \in A$	$(A, +)$ subgroup (char. of subgroups) + stable under $\cdot$; laws inherited.
Subfield $\Longleftrightarrow$ $0, 1 \in A$, $x - y \in A$, $xy^{-1} \in A$ $(y \neq 0)$	$(A, +) \leq (K, +)$, $(A^*, \cdot) \leq (K^*, \cdot)$; for $y \neq 0$: $xy = x(y^{-1})^{-1}$; $y = 0$: $xy = 0$.
Group hom.: $f(1) = 1'$, $f(x^{-1}) = f(x)^{-1}$	$f(1)f(1) = f(1)1'$, cancel; $f(x)f(x^{-1}) = f(1) = 1'$ both sides + uniqueness.
Ring hom. surjective $\Longrightarrow$ unitary; unitary $\Longrightarrow$ $f(x^{-1}) = f(x)^{-1}$	$x'f(1) = f(x)f(1) = f(x) = x' \Longrightarrow f(1)$ is an identity $\Longrightarrow = 1'$; $f(x)f(x^{-1}) = f(1) = 1'$.

Vector spaces, subspaces, linear maps (Courses 3–4)

Statement	Key steps

$k0 = 0v = 0$; $k(-v) = (-k)v = -kv$; $k(v - v')$; $(k - k')v$	apply (L1)/(L2) to a sum: $k0 + kv = k(0 + v) = kv \Longrightarrow$ cancel. Same pattern for all four.
$kv = 0 \Longleftrightarrow k = 0$ or $v = 0$	$k \neq 0$: $v = 1v = (k^{-1}k)v = k^{-1}(kv) = k^{-1}0 = 0$ (field!).
$S \leq V \Longleftrightarrow 0 \in S$, $k_1v_1 + k_2v_2 \in S$	$\Longrightarrow 0 = 0 \cdot v_1$; closure under $+$ and scalars. $\Longleftarrow k_1 = k_2 = 1$; $k_2 = 0$.
Intersection of subspaces is a subspace	0 in all; $k_1x + k_2y$ in each S_i.
$\langle X \rangle$ = linear combinations of X	L: (i) $L \leq V$, (ii) $X \subseteq L$, (iii) $L \subseteq S$ for every $S \supseteq X \Longrightarrow L$ is the smallest $\Longrightarrow L = \langle X \rangle$.
$S + T = \langle S \cup T \rangle$	$\subseteq$: $s + t = 1s + 1t$. $\supseteq$: split a combination into the S-part and the $(T \setminus S)$-part.
$V = S \oplus T \Longleftrightarrow$ unique $v = s + t$	$\Longrightarrow s - s' = t' - t \in S \cap T = \{0\}$. $\Longleftarrow 0 \neq v \in S \cap T$: $v = v + 0 = 0 + v$, two writings.
f linear $\Longleftrightarrow f(k_1v_1 + k_2v_2) = k_1f(v_1) + k_2f(v_2)$	$\Longleftarrow k_1 = k_2 = 1$; $k_2 = 0$.
f^{-1} linear; $g \circ f$ linear	$k_1v_1' + k_2v_2' = f(k_1v_1 + k_2v_2)$, apply f^{-1}; compose the linearity relations.
$\operatorname{Ker} f \leq V$, $\operatorname{Im} f \leq V'$	$f(0) = 0'$; $f(k_1v_1 + k_2v_2) = k_1 \cdot 0' + k_2 \cdot 0'$; $k_1f(v_1) + k_2f(v_2) = f(k_1v_1 + k_2v_2)$.
$\operatorname{Ker} f = \{0\} \Longleftrightarrow f$ injective	$\Longrightarrow f(v_1 - v_2) = 0' \Longrightarrow v_1 = v_2$. $\Longleftarrow f(v) = 0' = f(0) \Longrightarrow v = 0$.
$f(\langle X \rangle) = \langle f(X) \rangle$	$X = \emptyset$ separately; else $f(\sum k_iv_i) = \sum k_if(v_i)$.
$\operatorname{Hom}_K(V, V')$ is a vector space	$f + g$, kf linear (compute); subspace of $(V')^V$ with pointwise operations.

Independence, bases, dimension (Courses 5–6)

Statement	Key steps
Dependent $\Longleftrightarrow$ one vector is a combination of the others	$\Longrightarrow k_j \neq 0$, divide by $-k_j$. $\Longleftarrow (-1)v_j + \sum \alpha_iv_i = 0$.
n vectors of K^n dependent $\Longleftrightarrow \det = 0$	$\sum k_jv_j = 0$ is a square homogeneous system; non-zero solution $\Longleftrightarrow \det = 0$.
Every vector space has a basis	remove a vector that is a combination of the others while the list is dependent; generation is preserved; ends at an independent list.
Basis $\Longleftrightarrow$ unique writing	$\Longrightarrow \sum(k_i - k_i')v_i = 0 \Longrightarrow k_i = k_i'$. $\Longleftarrow$ unique writing of $0 \Longrightarrow$ independent.
Linear map determined by a basis; f, g agree on a basis $\Longrightarrow f = g$	$f(\sum k_iv_i) = \sum k_if(v_i)$.

f inj. + indep. $\Longrightarrow$ indep.; f surj. + gen. $\Longrightarrow$ gen.	$f(\sum k_i v_i) = 0' = f(0) \Longrightarrow \sum k_i v_i = 0$; $\langle f(X)\rangle = f(\langle X\rangle) = f(V) = V'$.
Steinitz: indep. $(x_1..x_m)$, gen. $(y_1..y_n) \Longrightarrow m \leq n$ + exchange	induction on m; $x_m = \sum_{i<m} k_i x_i + \sum_{i\geq m} k_i y_i$, some $k_j \neq 0$ with $j \geq m$ (independence), solve for y_j, replace it by x_m (Lemma: adding/removing a combination does not change $\langle\cdot\rangle$). If $m - 1 = n$: $x_m \in \langle x_1..x_{m-1}\rangle$, contradiction.
All bases have the same size	Steinitz both ways: $m \leq n$, $n \leq m$.
$\dim V = n \Longleftrightarrow$ max. indep. $= n \Longleftrightarrow$ min. gen. $= n$	Steinitz each time (independent list $\leq$ generating list).
$n = \dim V$ vectors: indep. $\Longleftrightarrow$ generating	$\Longrightarrow$ Steinitz replaces all of a basis. $\Longleftarrow$ dependent $\Longrightarrow n-1$ generators, contradiction.
Completion to a basis	Steinitz: $(u_1..u_m, v_{m+1}..v_n)$ generates, has $\dim V$ vectors $\Longrightarrow$ basis.
$S \leq V$: $\dim S \leq \dim V$; $= \Longleftrightarrow S = V$	extend a basis of S; if $\dim S = \dim V$, a basis of S is a basis of V.
Every subspace has a complement	complete a basis of S with $v_{m+1}..v_n$, $\bar{S} = \langle v_{m+1}..v_n\rangle$; sum by writing in the basis, intersection $\{0\}$ by independence.
$V \simeq V' \Longleftrightarrow \dim V = \dim V'$	$\Longrightarrow$ iso sends a basis to a basis. $\Longleftarrow f(\sum k_i v_i) = \sum k_i v_i'$, linear and bijective (unique coordinates).
First Dimension Th. $\dim V = \dim \operatorname{Ker} f + \dim \operatorname{Im} f$	basis $u_1..u_m$ of $\operatorname{Ker} f$, complete with $v_{m+1}..v_n$; show $f(v_{m+1})..f(v_n)$ is a basis of $\operatorname{Im} f$ (indep.: combination in $\operatorname{Ker} f \Longrightarrow$ all coeffs 0; gen.: $f(u_i) = 0$).
$\dim V = \dim V'$: injective $\Longleftrightarrow$ surjective	First Dimension Theorem + "subspace of full dimension = whole space".
Second Dimension Th. $\dim S + \dim T = \dim(S \cap T) + \dim(S + T)$	$f : S \times T \to S + T$, $f(s,t) = s + t$: $\dim(S \times T) = \dim S + \dim T$, $\operatorname{Im} f = S + T$, $\operatorname{Ker} f = \{(s,-s) \mid s \in S \cap T\} \simeq S \cap T$; apply the First Dimension Theorem.

Definitions you must state exactly

operation, stable subset, semigroup / monoid / group, ring / field, subgroup / subring / subfield, group and ring homomorphism, relation, function, injective / surjective / bijective, equivalence relation, partition, quotient set, vector space (with (L1)–(L4)), subspace, generated subspace, linear combination, finitely generated, sum and direct sum, linear map / isomorphism / endomorphism / automorphism, kernel and image, linear (in)dependence, basis, coordinates, dimension, complement, nullity and rank of a linear map.

> **Common mistakes in exam proofs.** (1) Proving a subspace property for specific vectors instead of arbitrary ones. (2) Forgetting $0 \in S$ (or $S \neq \emptyset$). (3) Using "dim" arguments before proving that the vectors are independent. (4) In Steinitz, assuming the *first* vector of Y can be exchanged — you must pick one with a non-zero coefficient. (5) For a function defined on classes ($\widehat{k} \mapsto \ldots$), forgetting to check it is well defined. (6) Saying "$f(0) = 0$, so f is linear" — this is only a necessary condition.